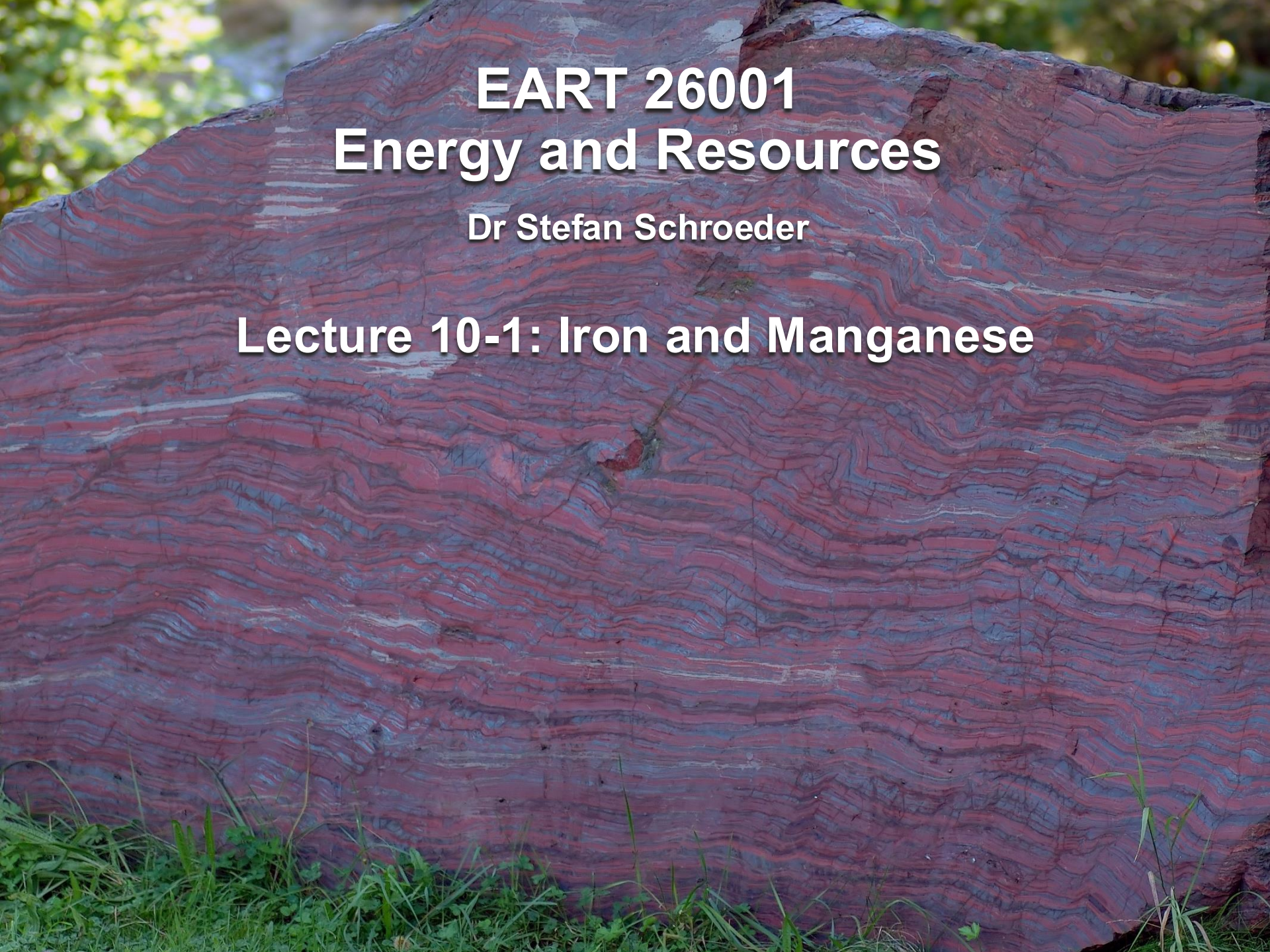


EART 26001
Energy and Resources

Dr Stefan Schroeder

Lecture 10-1: Iron and Manganese

Energy and Resources
Learning goals week 10

This week you will learn to:

- **Describe the uses and physical and chemical properties of iron and manganese, and how they contribute to formation of iron and manganese deposits.**
- **Recognize the sedimentary geological processes forming marine chemical iron and manganese deposits, and their modifications via supergene surface processes.**
- **Recognize the environmental impact of iron and manganese mining.**

Iron and Manganese
Uses

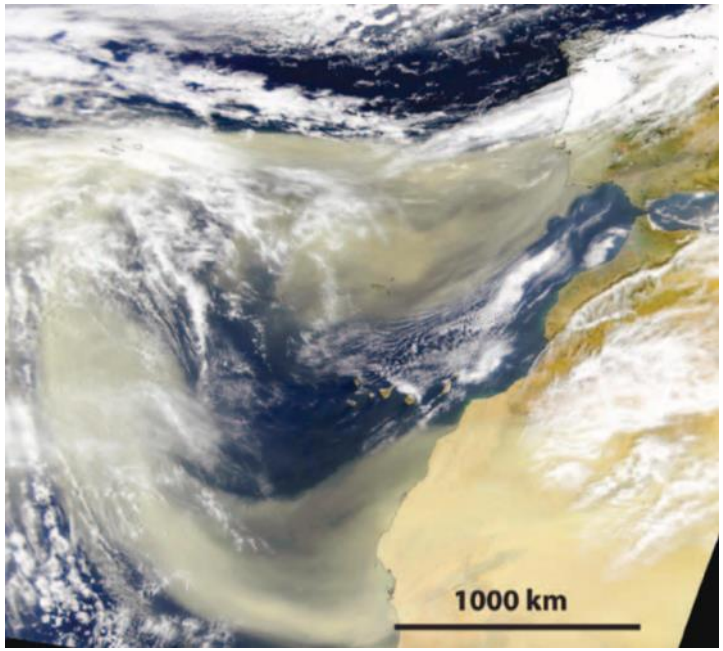




© Encyclopedia Britannica



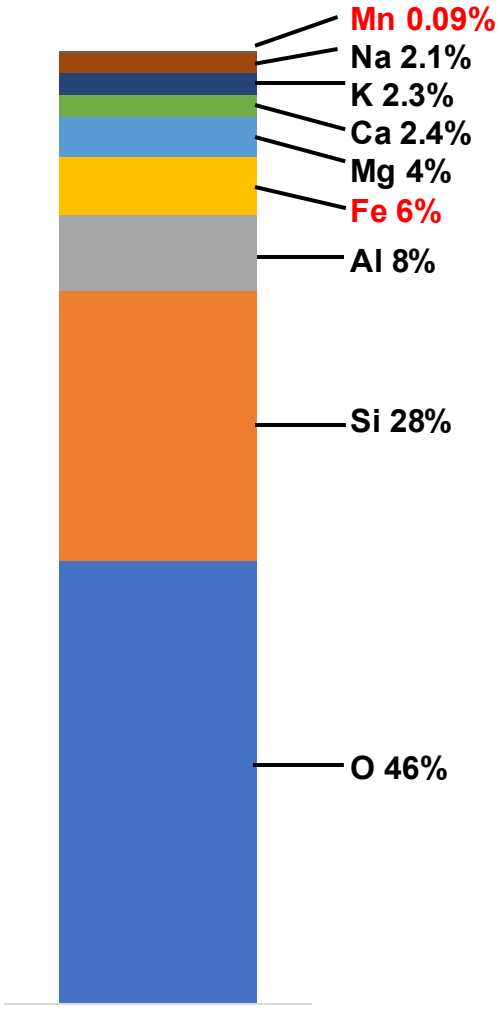
© 911metallurgist



Taylor 2011

Iron and Manganese
Properties

Concentration of major elements in crust



- **Enrichment to ore grade:**
  - **Fe ~5x**
  - **Mn ~220x**

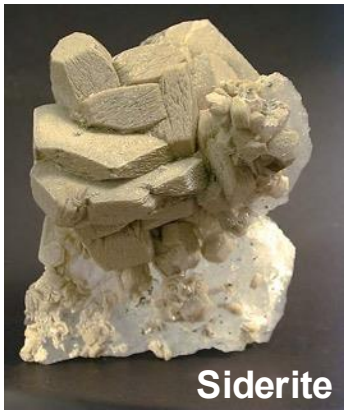
Key properties

	Iron	Manganese
Symbol	Fe	Mn
Atomic number	26	25
Atomic mass	55.845	54.938
Melting point	1538°C	1246°C
Boiling point	2861°C	2061°C
Density	7.87 g/cm^3	7.3 g/cm^3
Hardness	4	6

Iron and Manganese
Iron minerals

FERROUS (Fe^{2+}):
relatively **soluble** in reducing fluids

- Siderite ($FeCO_3$)
- Chamosite ($Fe(Si,Al)_2O_5(OH)_4$)
- Pyrite (FeS_2)



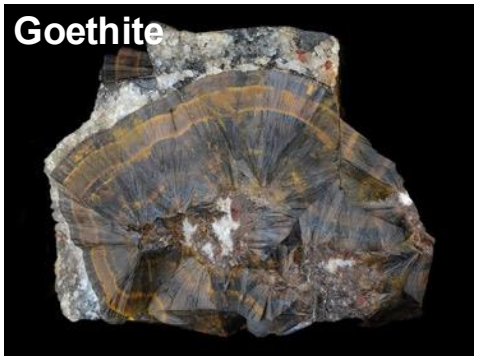




FERRIC (Fe^{3+}):
relatively **insoluble** in oxidizing fluids

- Hematite (Fe_2O_3)
- Goethite (FeOOH)
- Limonite (FeOOH H_2O)

- Magnetite (Fe_3O_4)

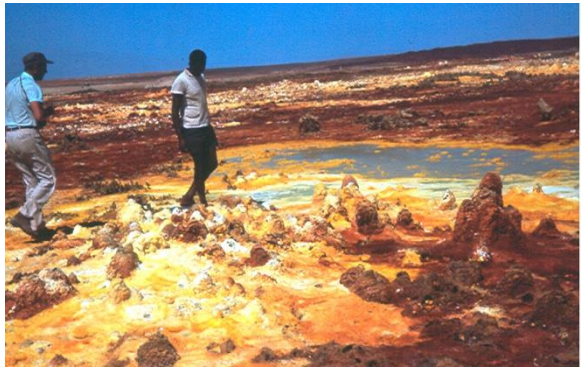


© Royal Cornwall Museum





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© C Schreiber

Iron and Manganese
Manganese minerals

MANGANOUS (Mn^{2+}):
relatively **soluble** in reducing fluids

- Rhodochrosite ($MnCO_3$)
- Kutnahorite ($CaMn(CO_3)_2$)



© mindat.org



© J Gutzmer, Freiberg

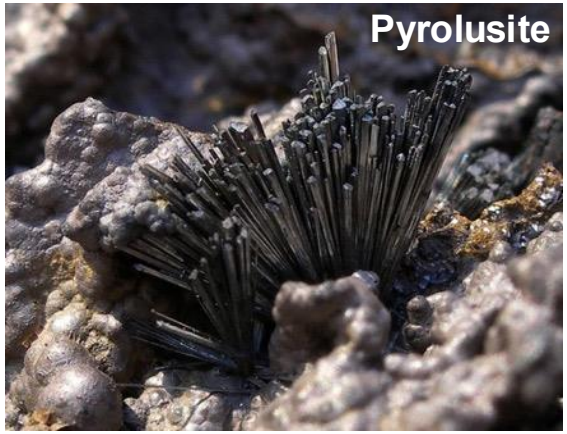
MANGANIC (Mn^{3+}/ Mn^{4+}):
relatively **insoluble** in oxidizing fluids

- Braunite ($Mn^{2+}Mn^{3+}{}_6(SiO_4)O_8$)
- Hausmannite ($Mn^{2+}Mn^{3+}{}_2O_4$)

- Pyrolusite ($Mn^{4+}O_2$)



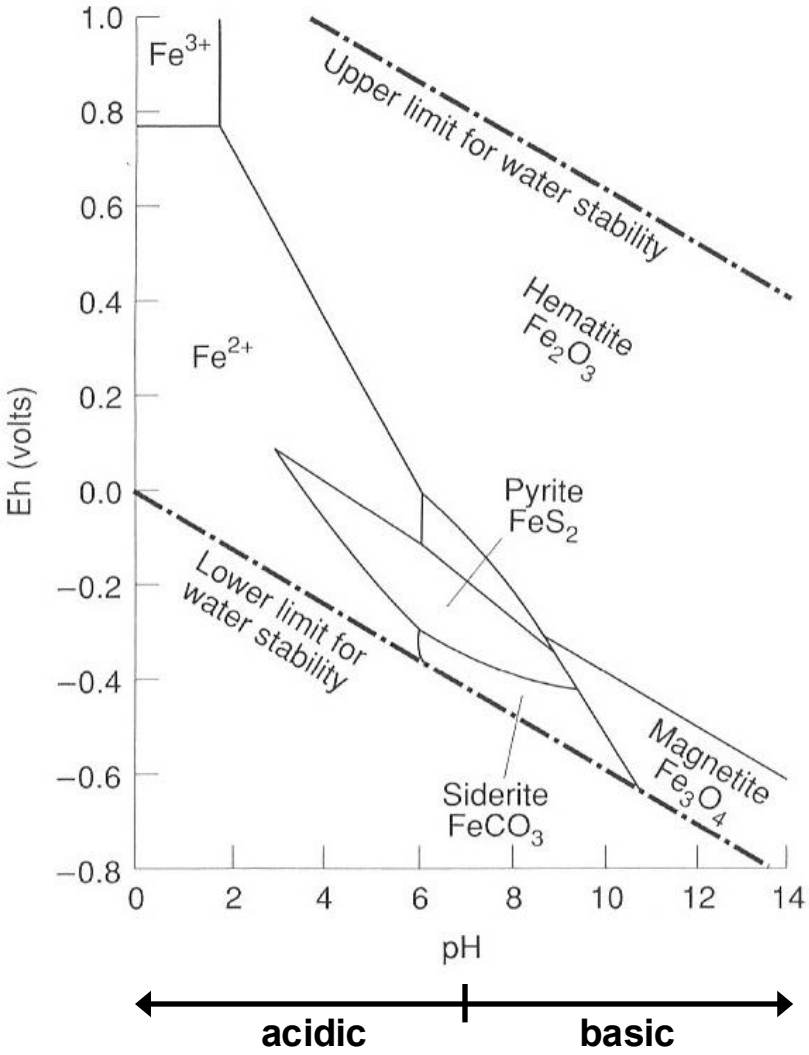
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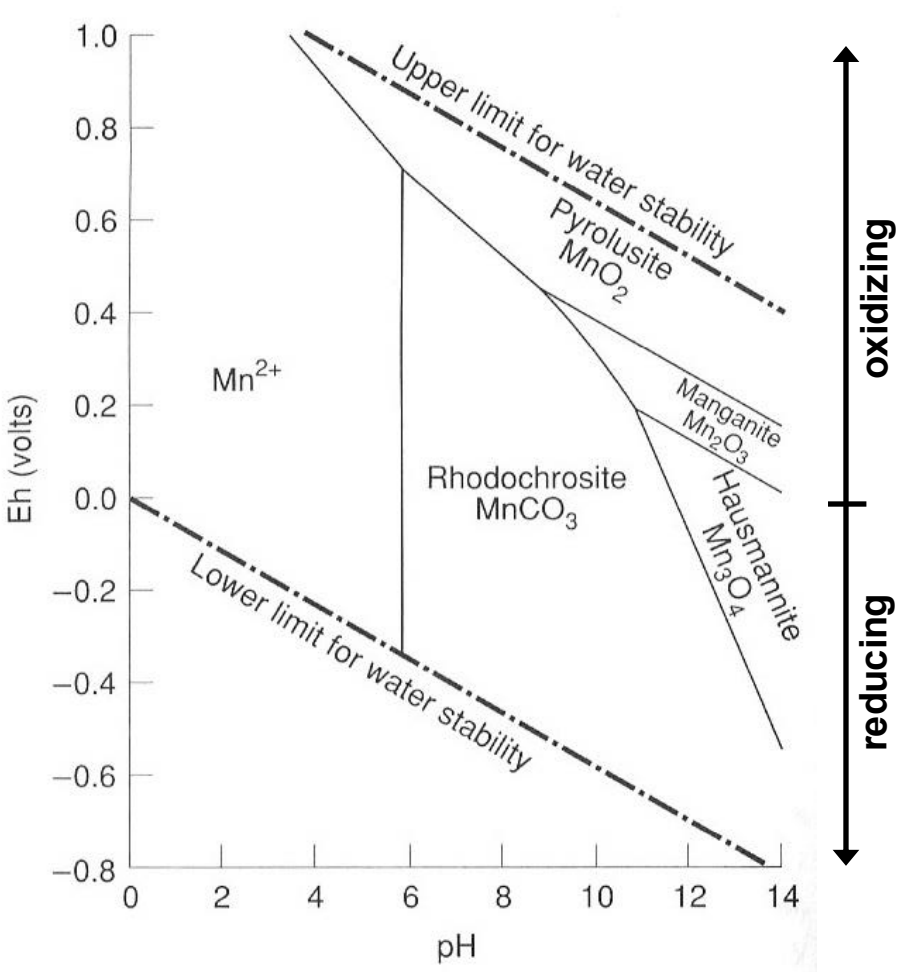
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Iron and Manganese
Redox chemistry

Fe



Mn

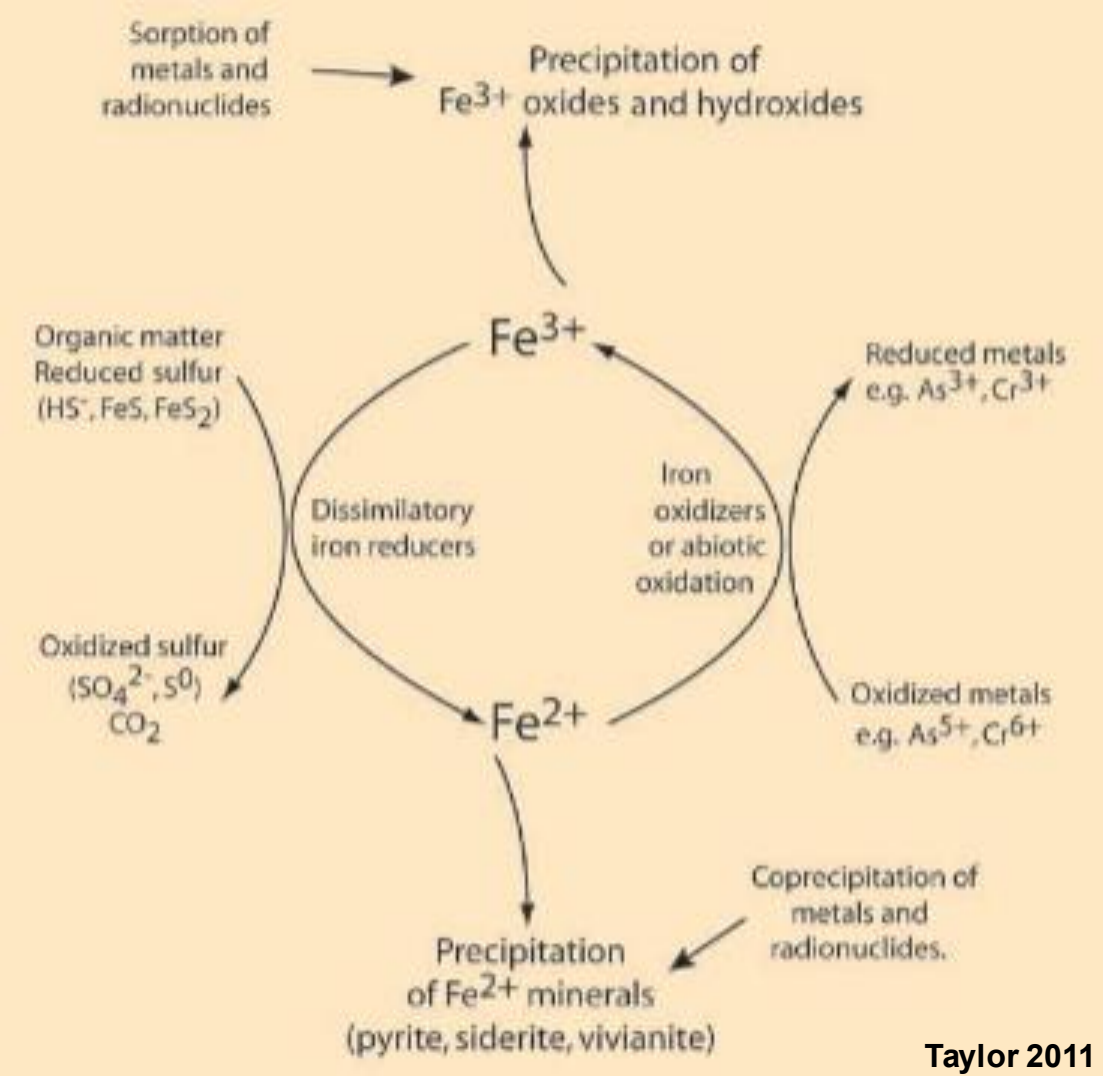


Robb 2005

$2Fe^{2+} + 3CO_2 <-> Fe^{3+}{}_2O_3 + 3CO$

$2Mn^{2+} + O_2 + 2H_2O <-> 2Mn^{4+}O_2 + 4H^+$

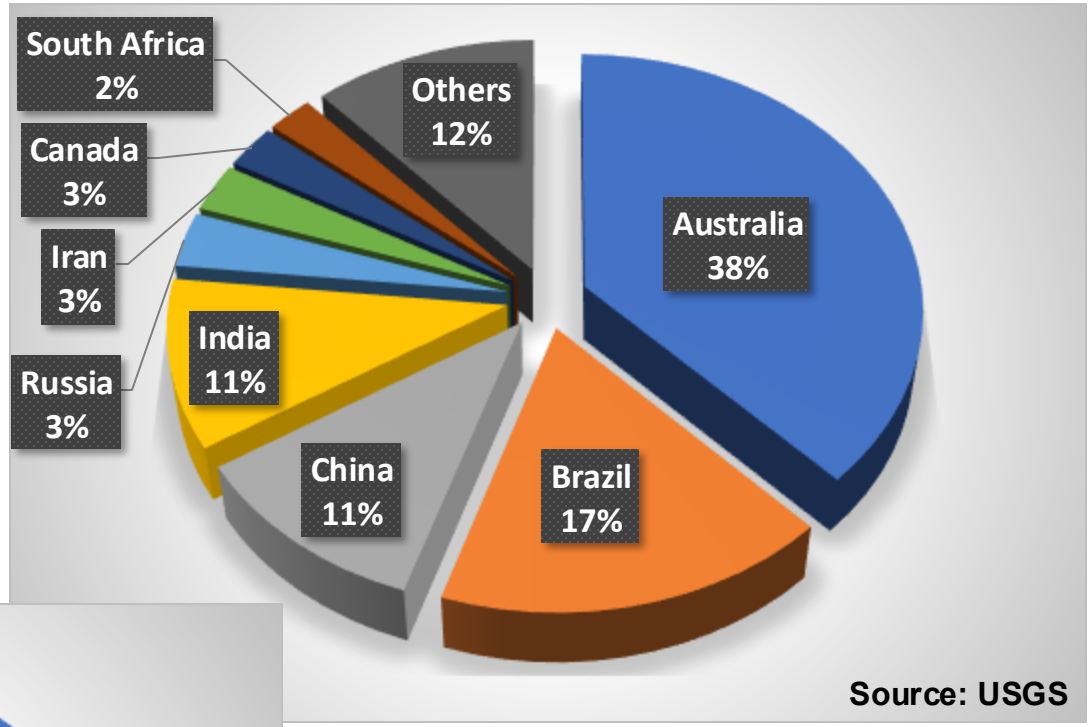
Iron and Manganese
Biochemical cycling



Taylor 2011

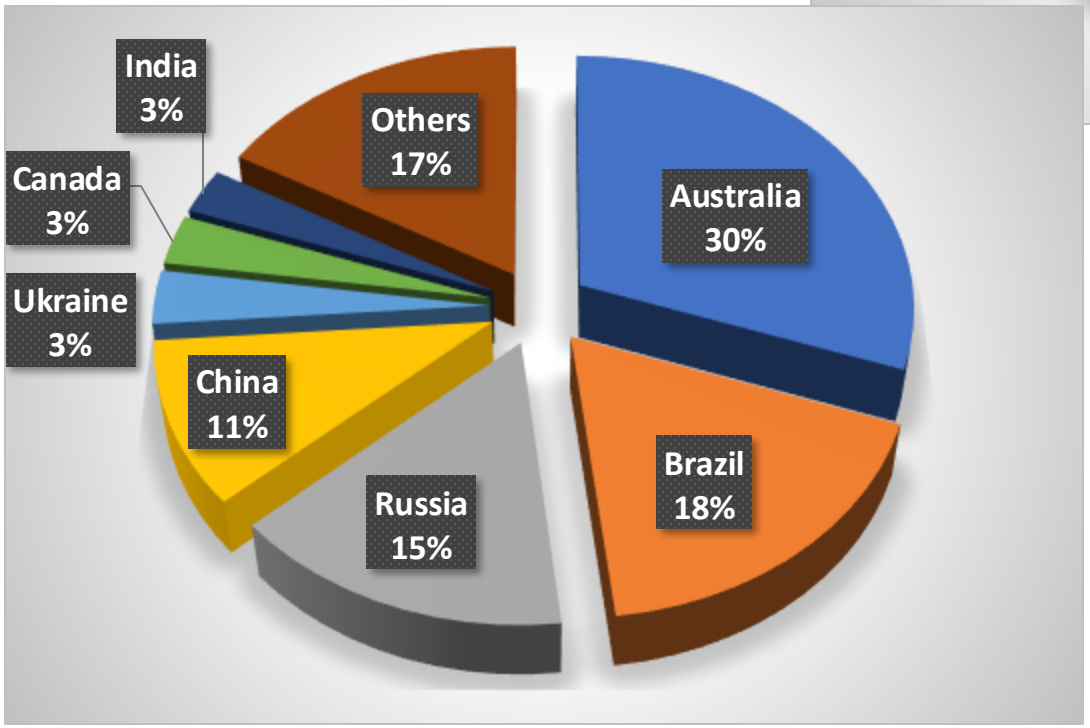
Iron and Manganese
Reserves - Fe

Fe ore Mine production 2023



Source: USGS

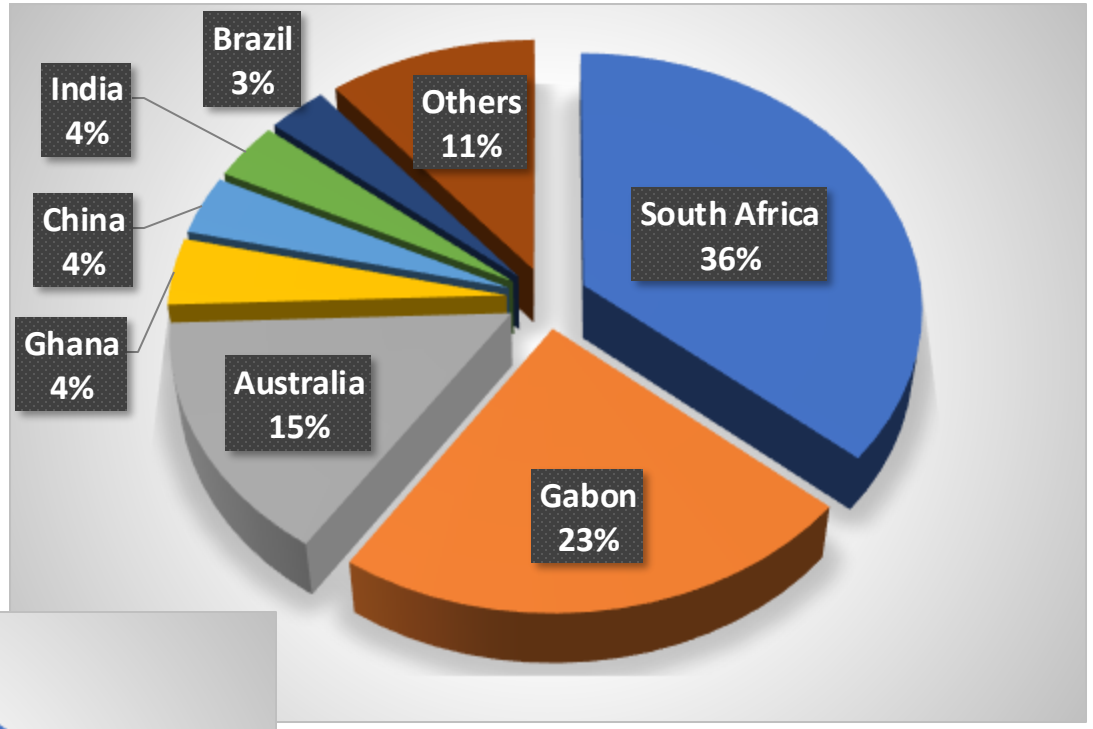
Fe Reserves (by raw ore) 2023



Source: USGS

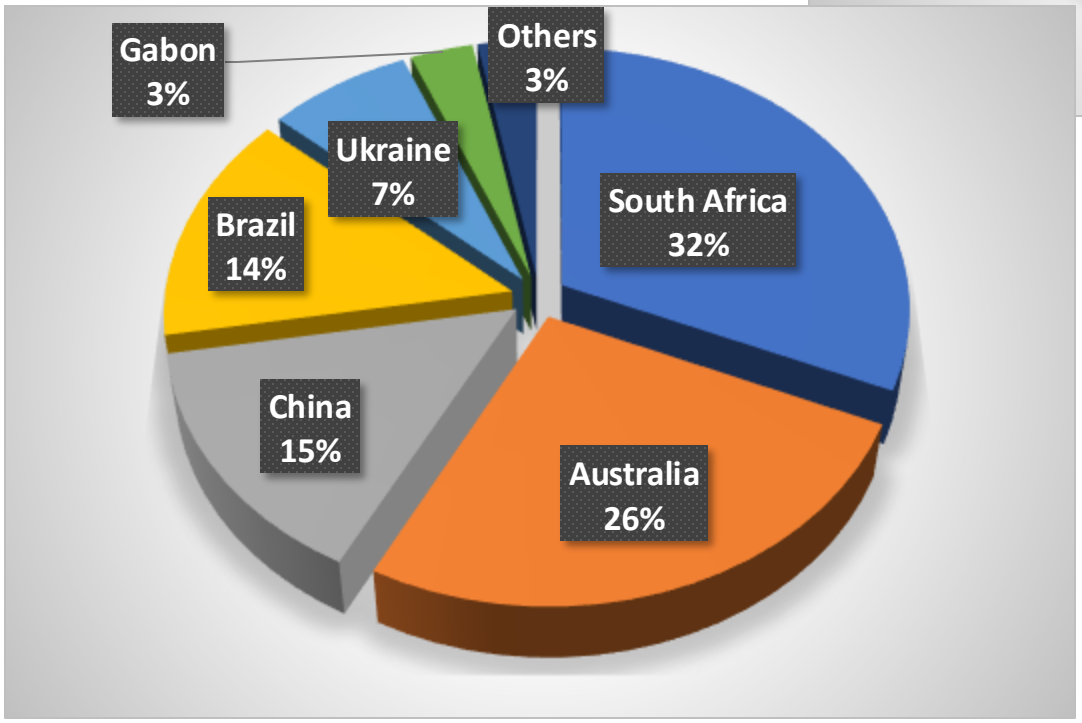
Iron and Manganese
Reserves - Mn

Mn Mine production 2023



Source: USGS

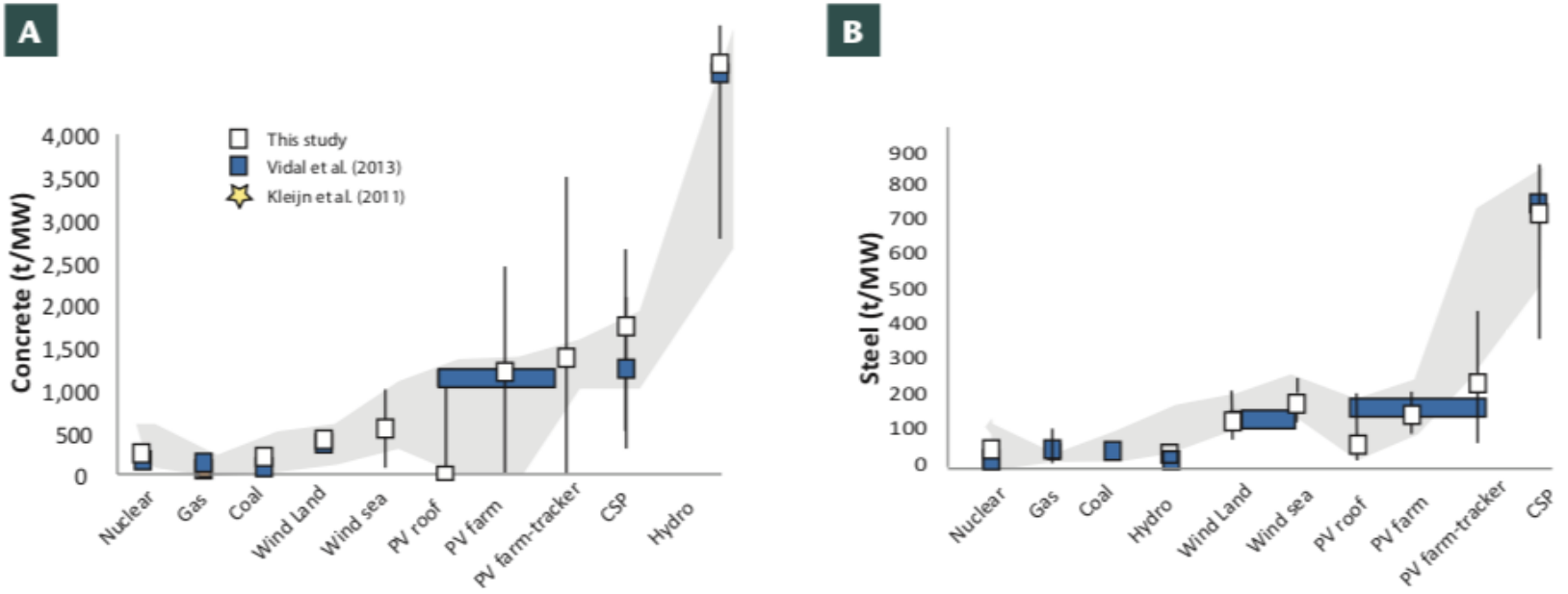
Mn Reserves 2023



Source: USGS

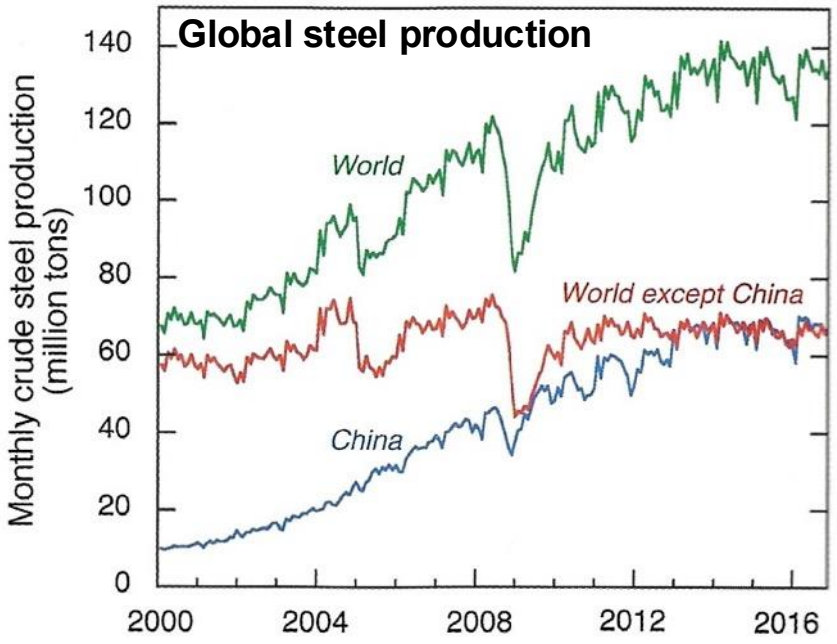
Iron and Manganese
Uses and demand

Resource need by energy generated for various energy sources

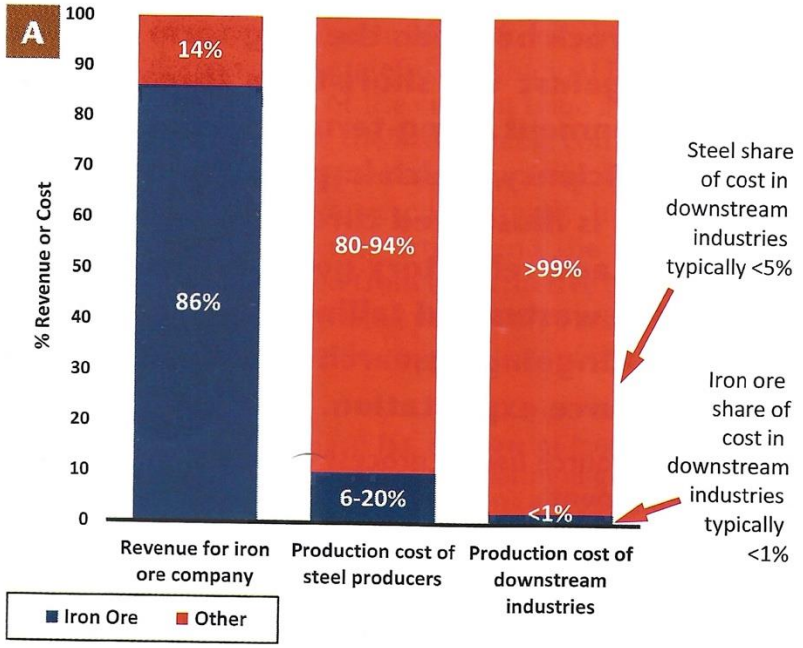


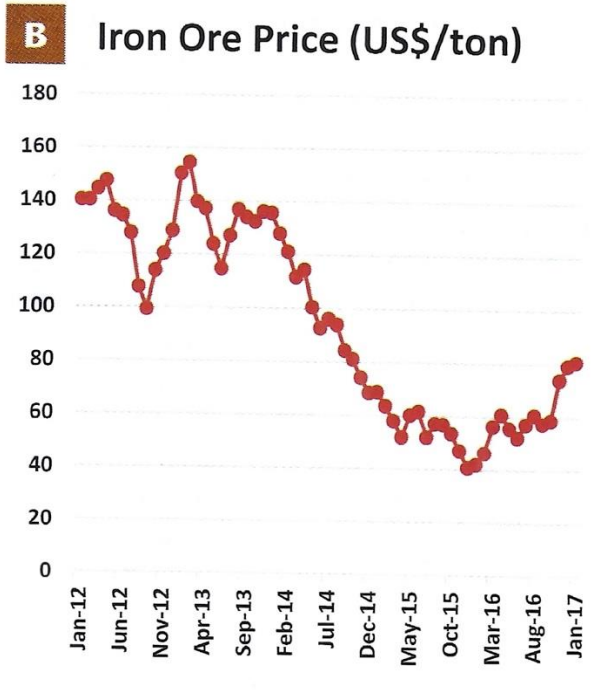
Vidal 2017

Iron and Manganese
Fe market



Calas (2017)

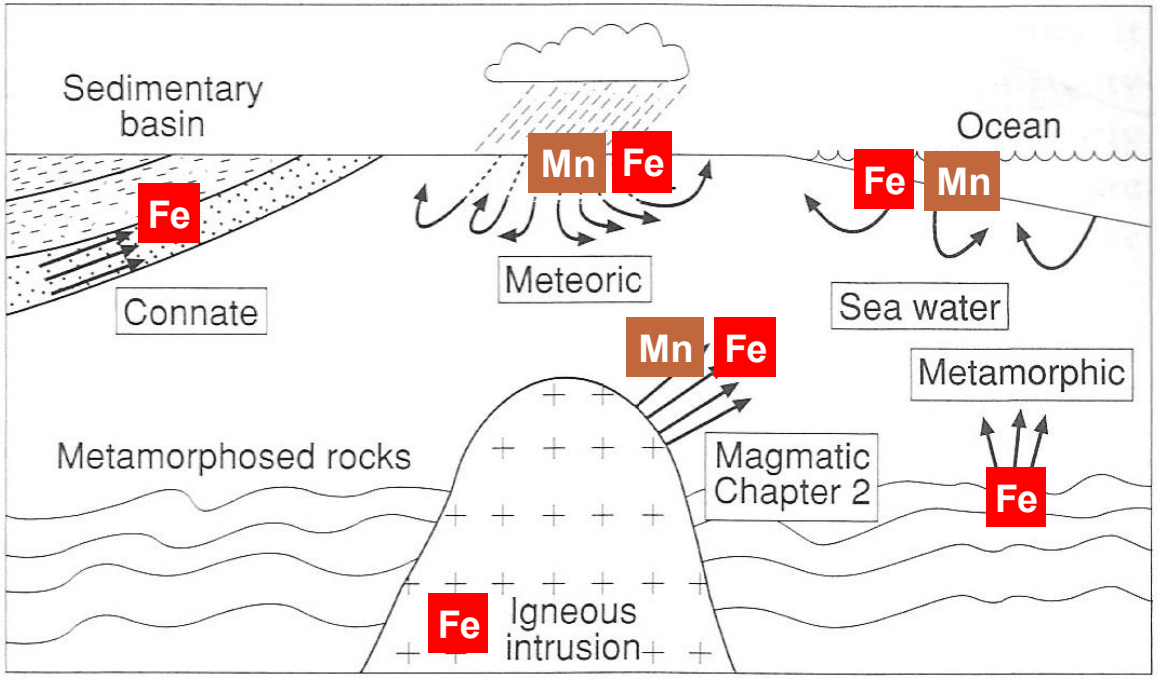




De Villiers (2017)

Iron and Manganese
Deposit types

Fluid/Deposit	Fe	Mn
Magma	✓	✗
Volcanic exhalations	✓	✓
Metamorphic fluids	✓	✗
Hydrothermal veins, replacements, disseminations	✓	✓
Marine chemical sediments	✓	✓
Meteoric fluids, surface weathering	✓	✓



Robb 2008

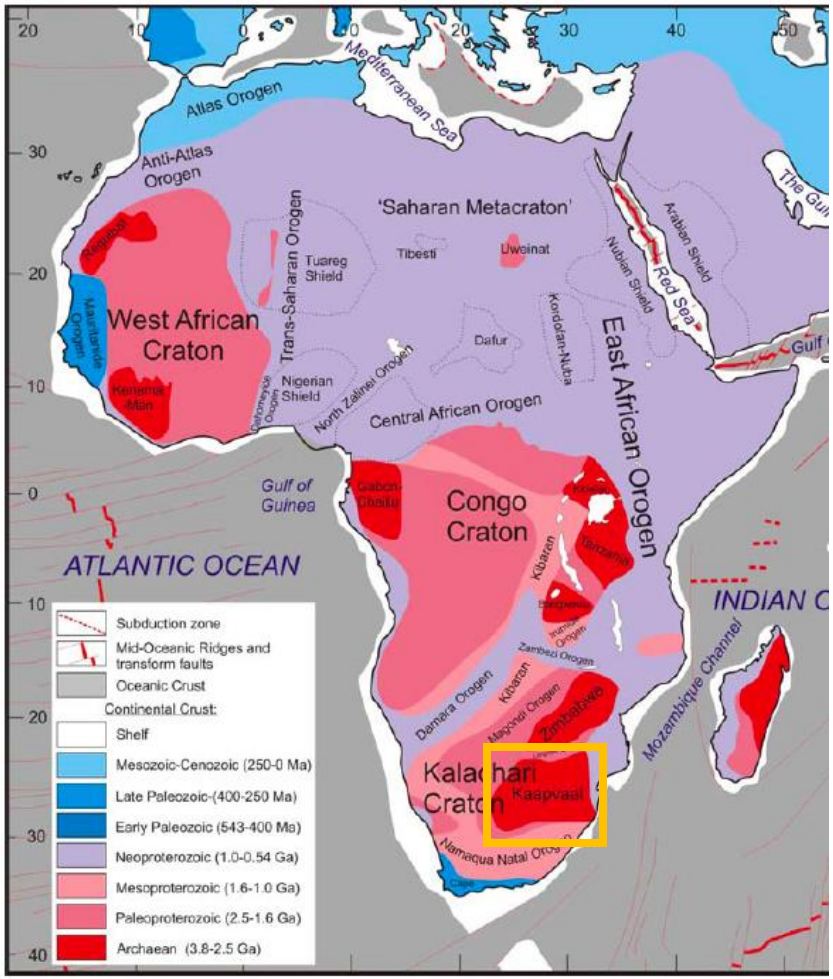
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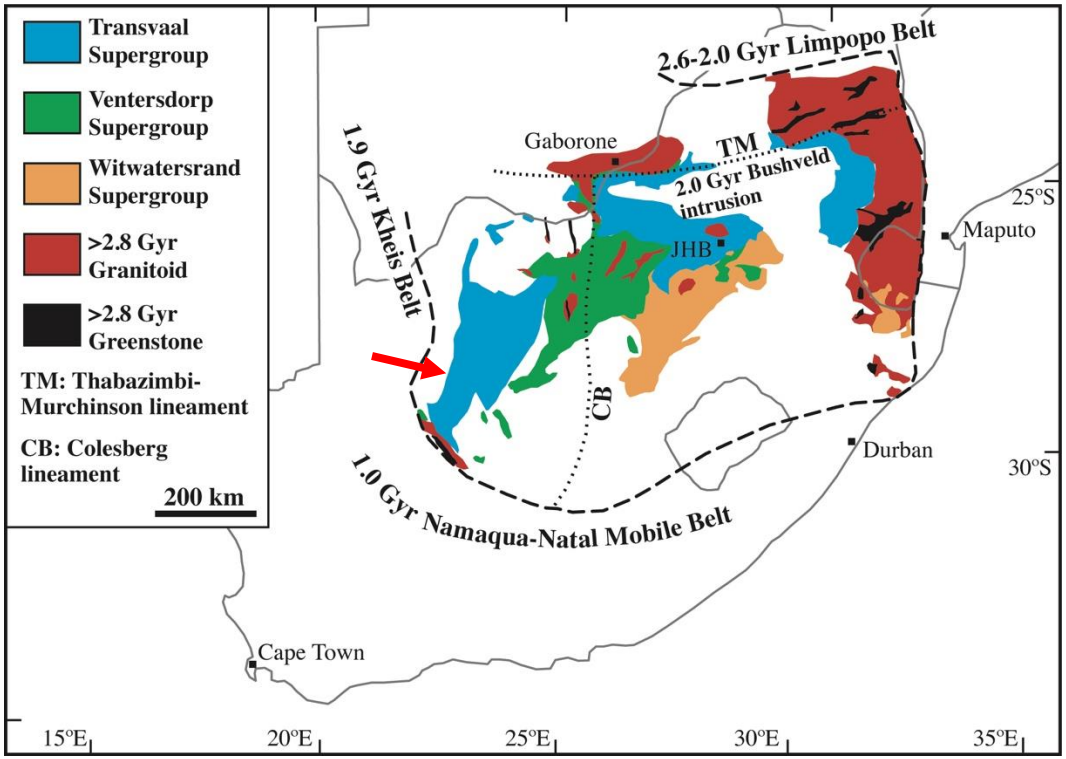
Lecture 10-2: Iron and manganese formations (I): deposition



Banded iron formations
Stratigraphy

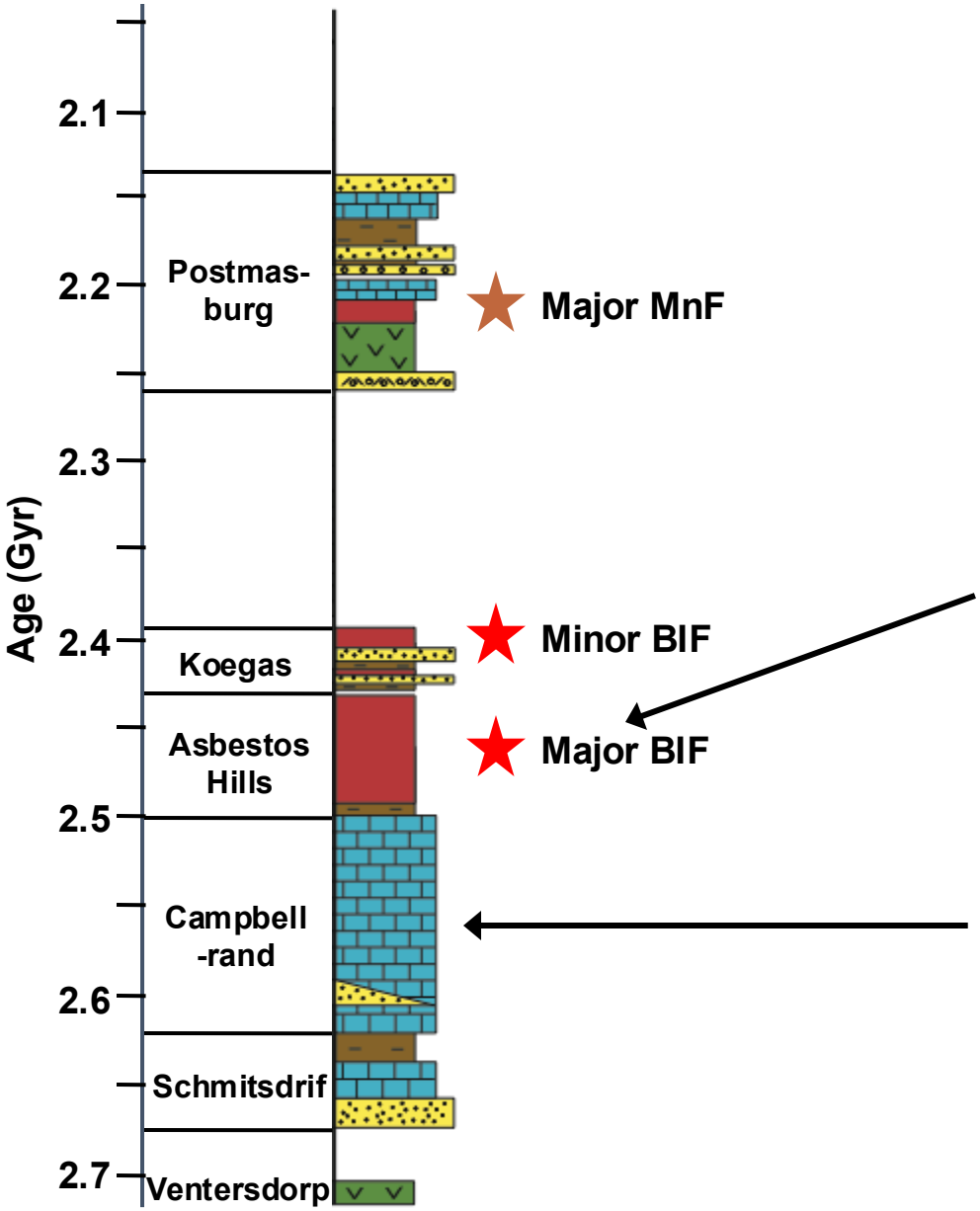


Van Hinsbergen 2011, after Gubanov & Mooney 2009

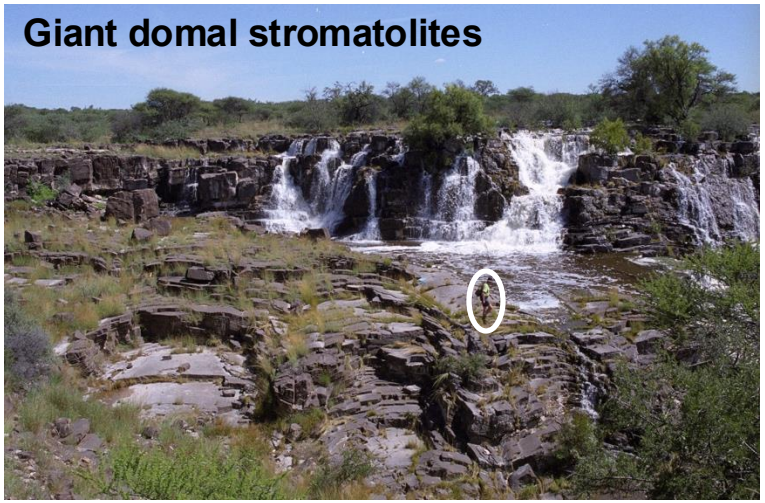


after Schmitz 2004

Banded iron formations
Stratigraphy

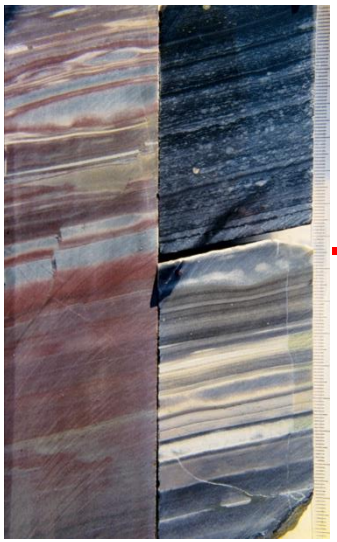


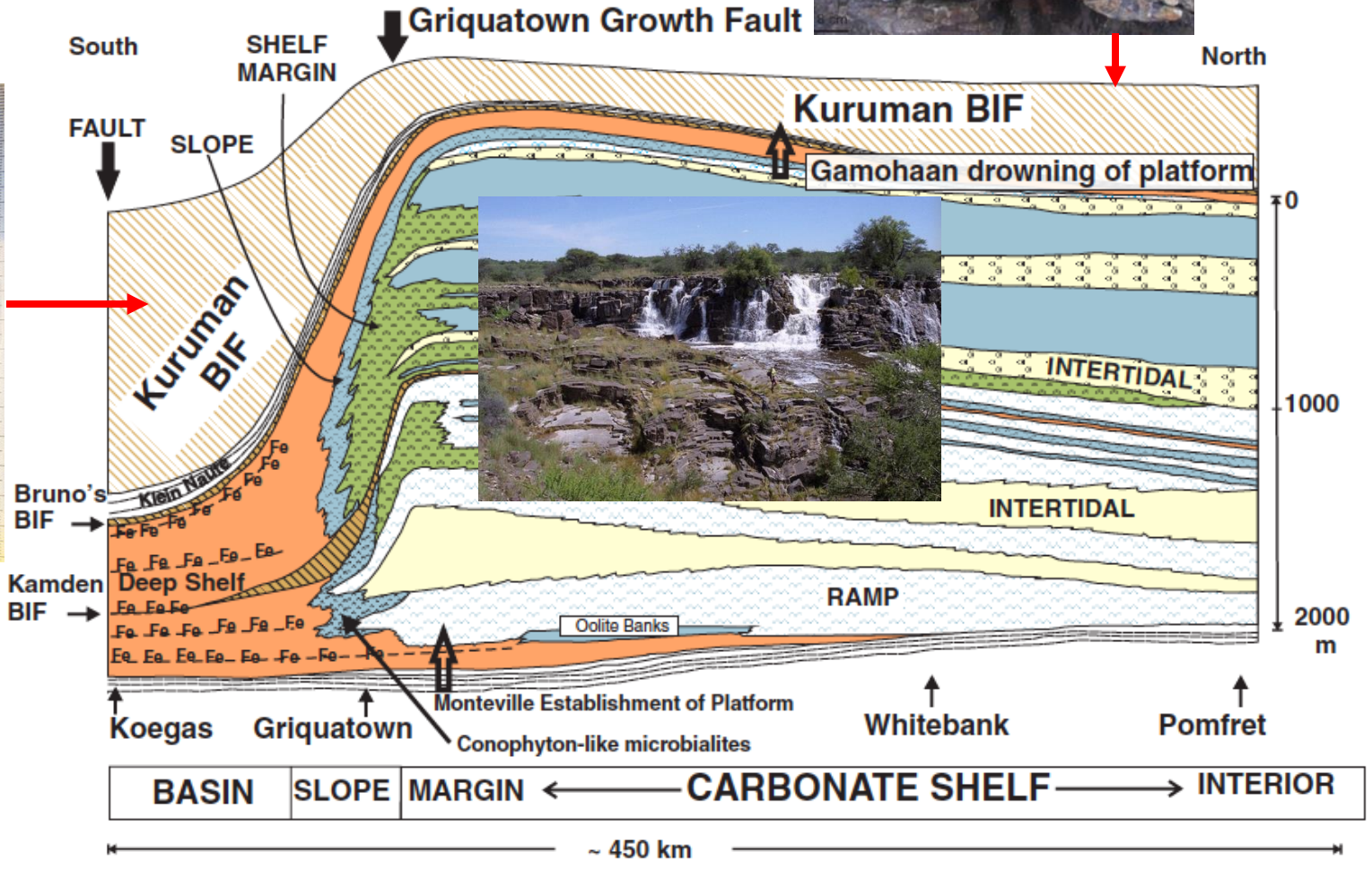




Banded iron formations
Sedimentology

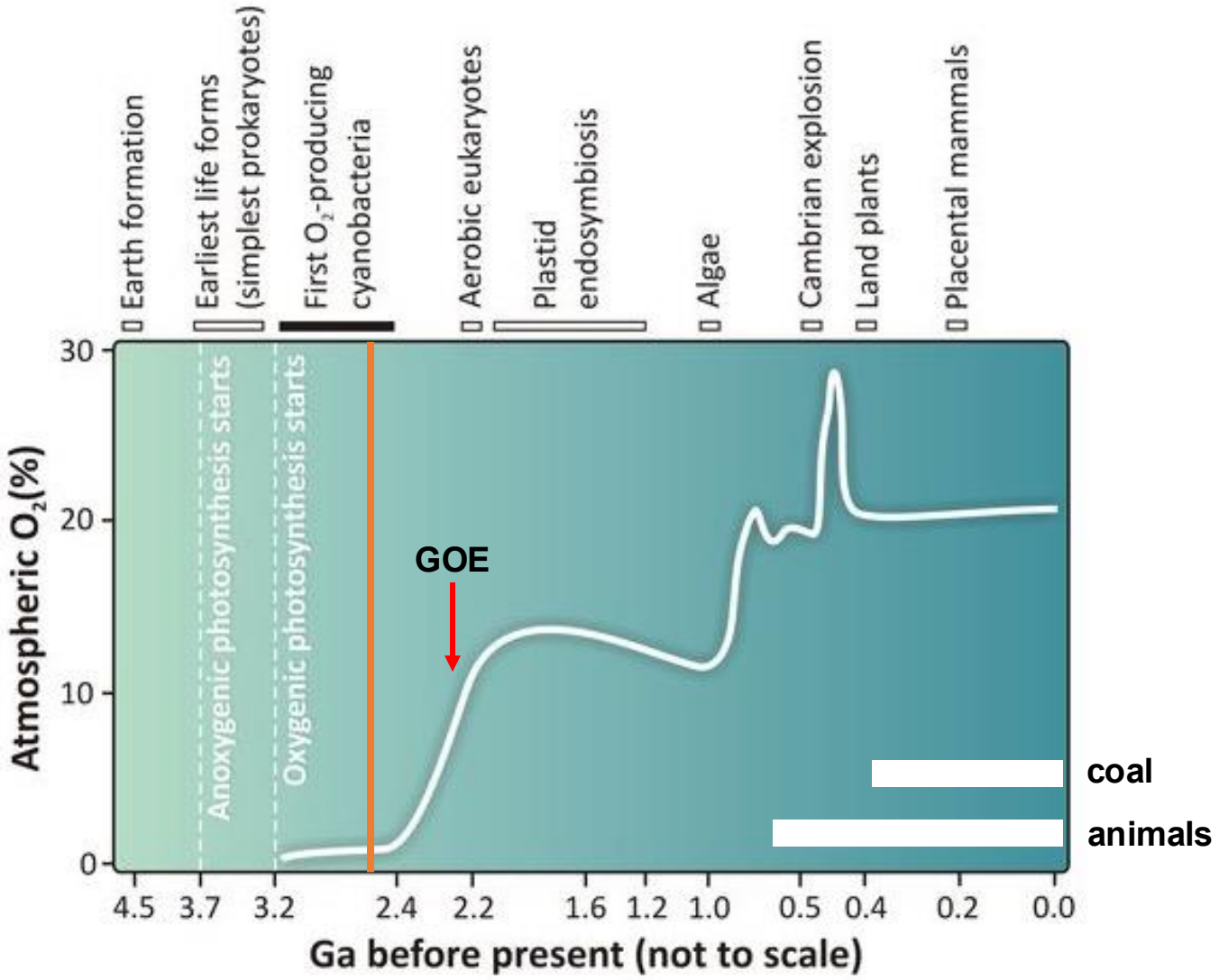






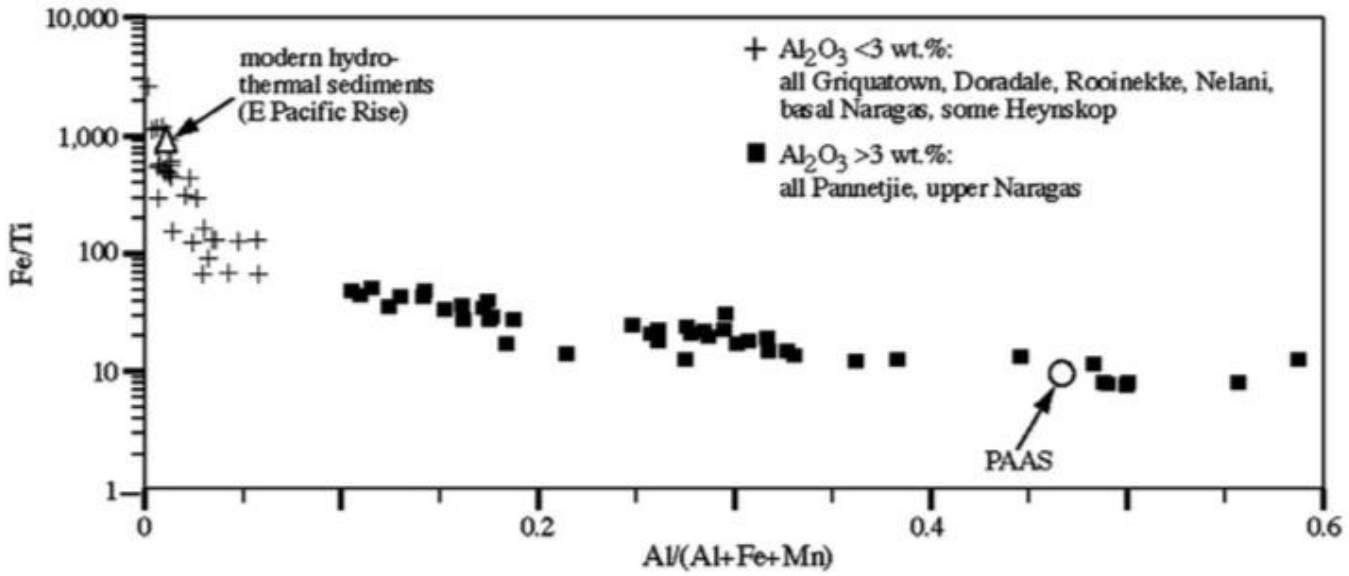
Beukes 2008

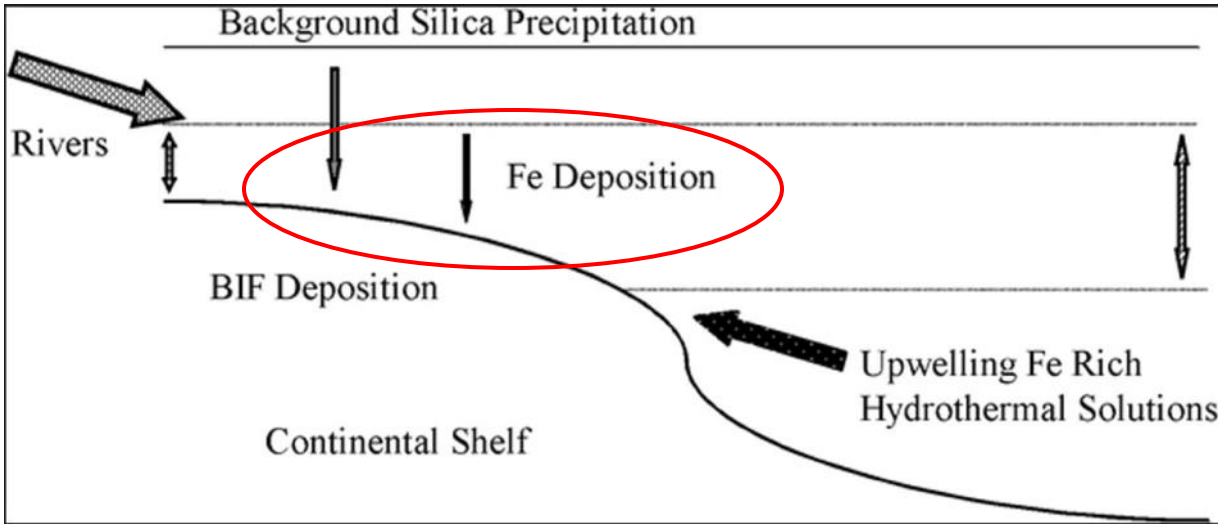
Banded iron formations
Implications for atmospheric oxygen



© NASA Astrobiology

Banded iron formations
Depositional processes





- Original deposits were Fe-silicate gels, Fe carbonates, and silica gels
- Fe occurs as hematite, siderite +/- magnetite
- <30% iron content

Perry 2007

Banded iron formations
Depositional processes

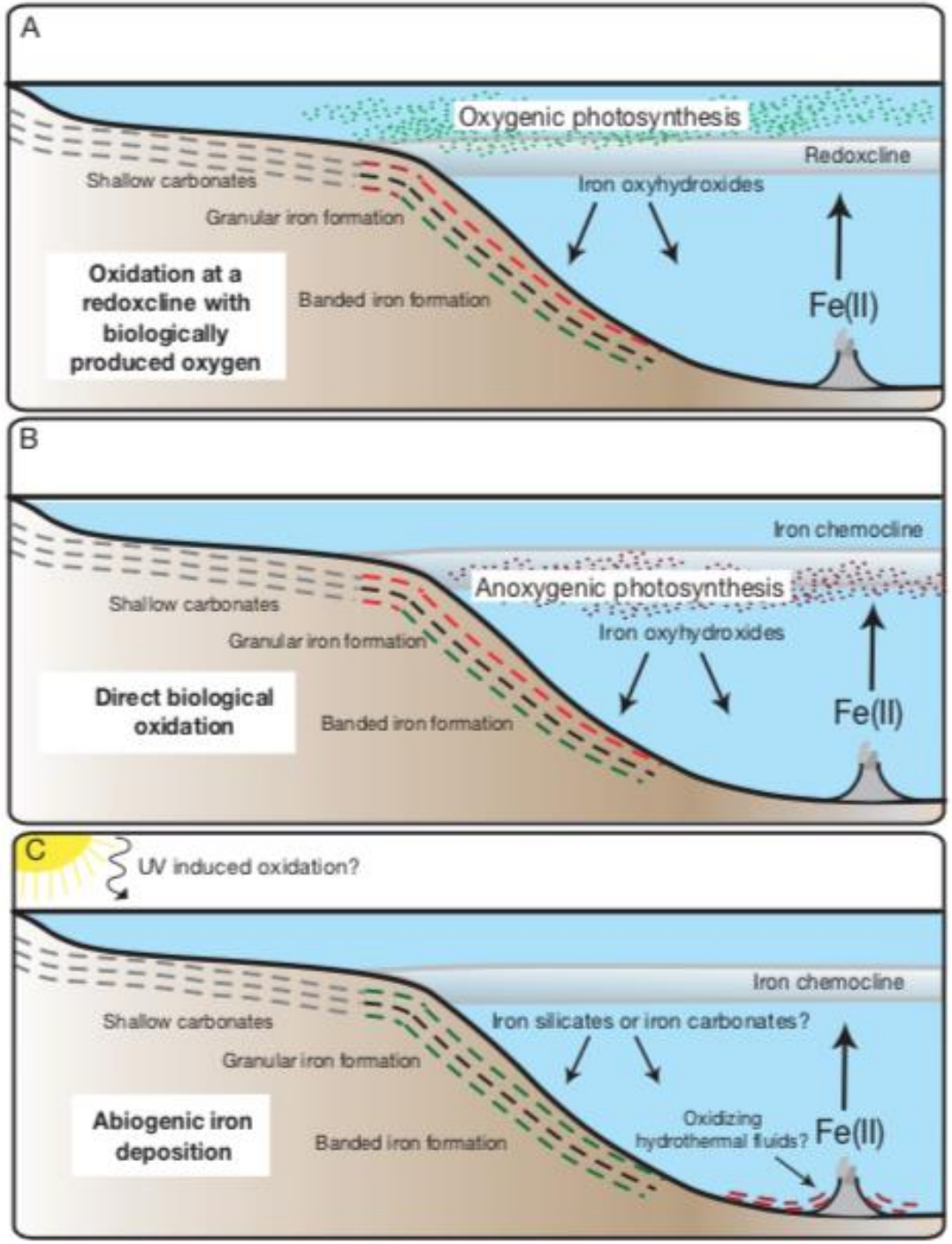
$2Fe^{2+} + 0.5O_2 + 5H_2O \rightarrow 2Fe^{3+}(OH)_3 + 4H^+$

Micro-aerobic oxidation:
$6Fe^{2+} + 0.5O_2 + CO_2 + 16H_2O \rightarrow CH_2O + 6Fe^{3+}(OH)_3 + 12H^+$ (Fe bacteria)

Anaerobic oxidation:
$10Fe^{2+} + 2NO_3^- + 24H_2O \rightarrow 10Fe^{3+}(OH)_3 + N_2 + 18H^+$

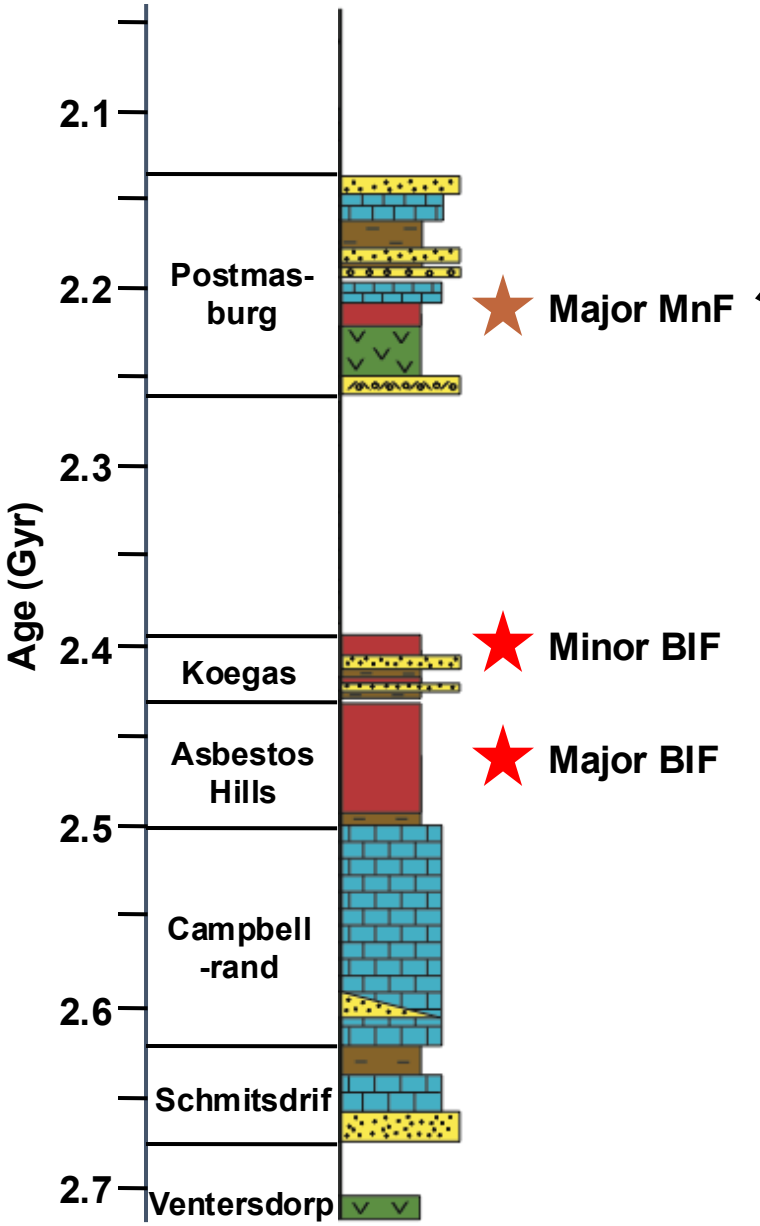
Anoxygenic photosynthesis:
$4Fe^{2+} + HCO_3^- + 10H_2O + hv \rightarrow CH_2O + 4Fe^{3+}(OH)_3 + 7H^+$ (purple/green bacteria)

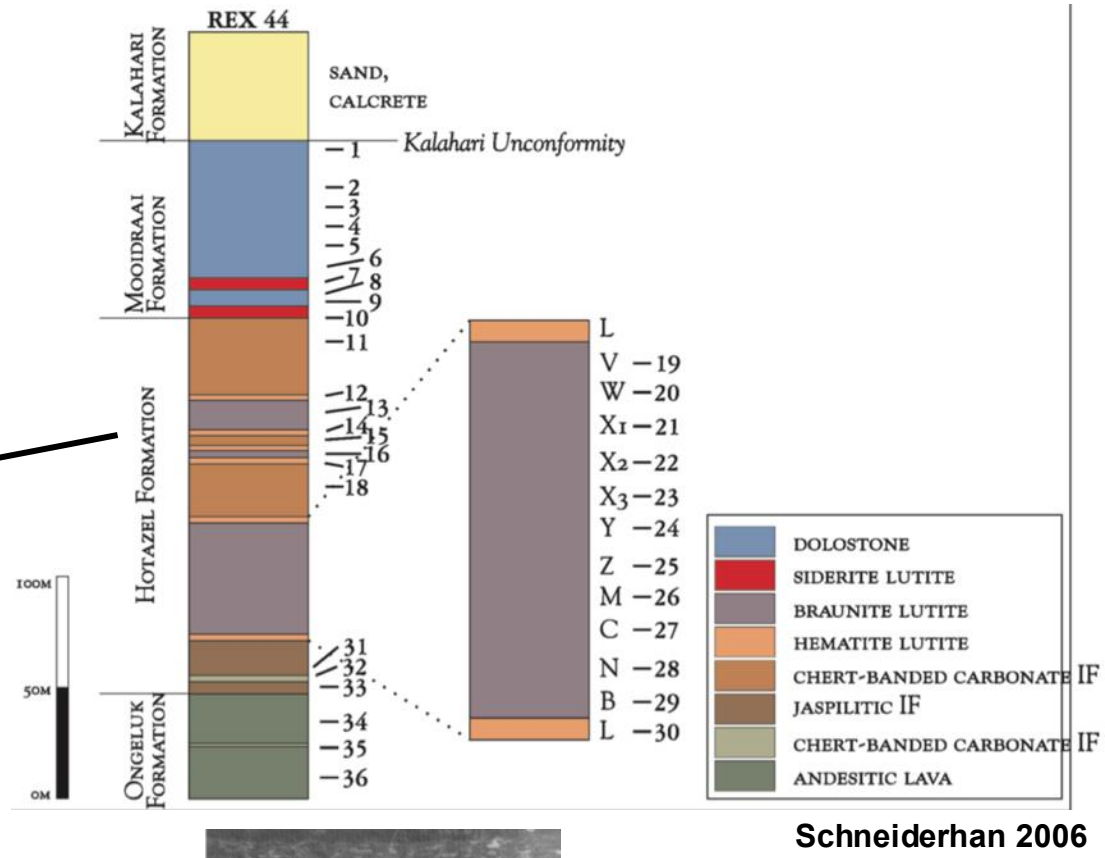
$2Fe^{2+} + 2H^+ + hv \rightarrow 2Fe^{3+} + H_2$



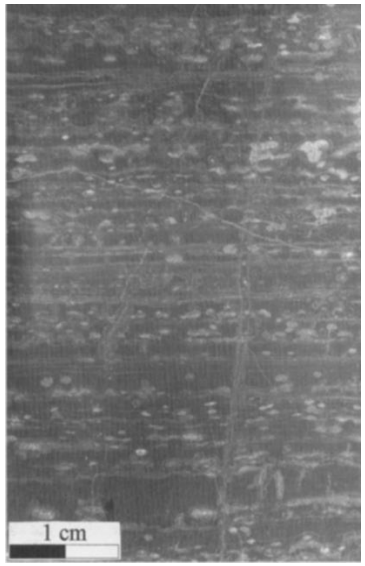
Bekker 2010

Manganese formations
Stratigraphy



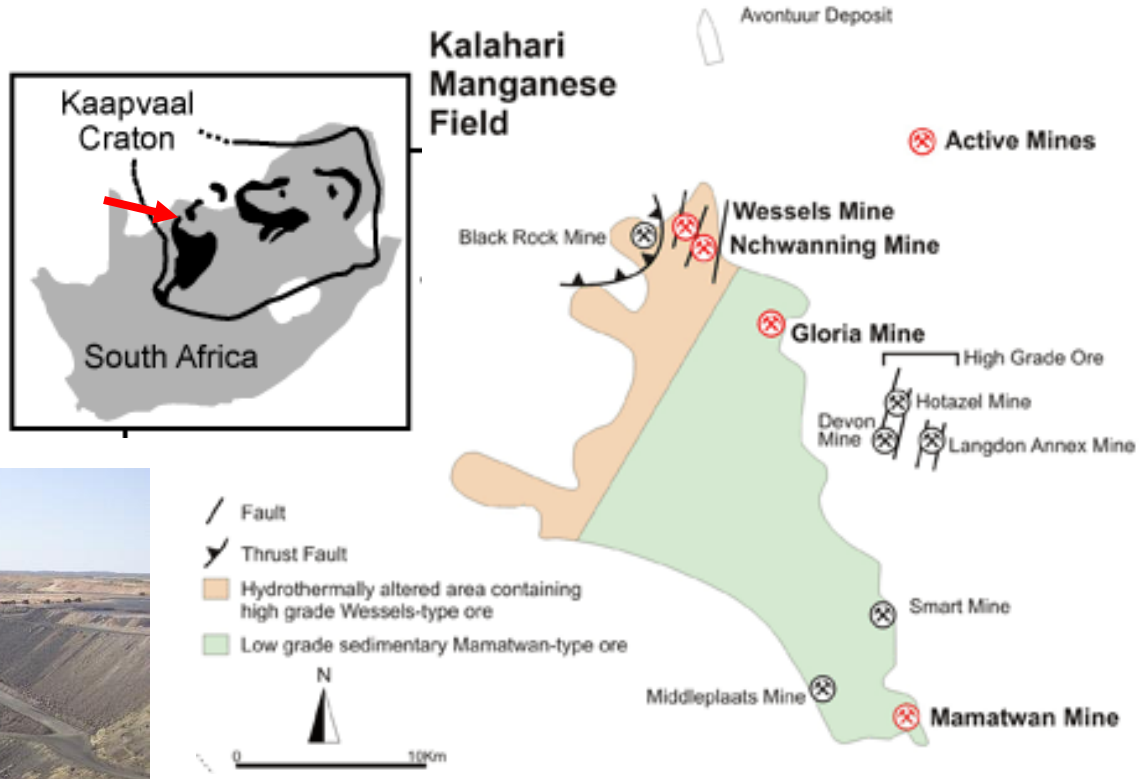


Schneiderhan 2006



Beukes 1996

Manganese formations
Kalahari manganese field



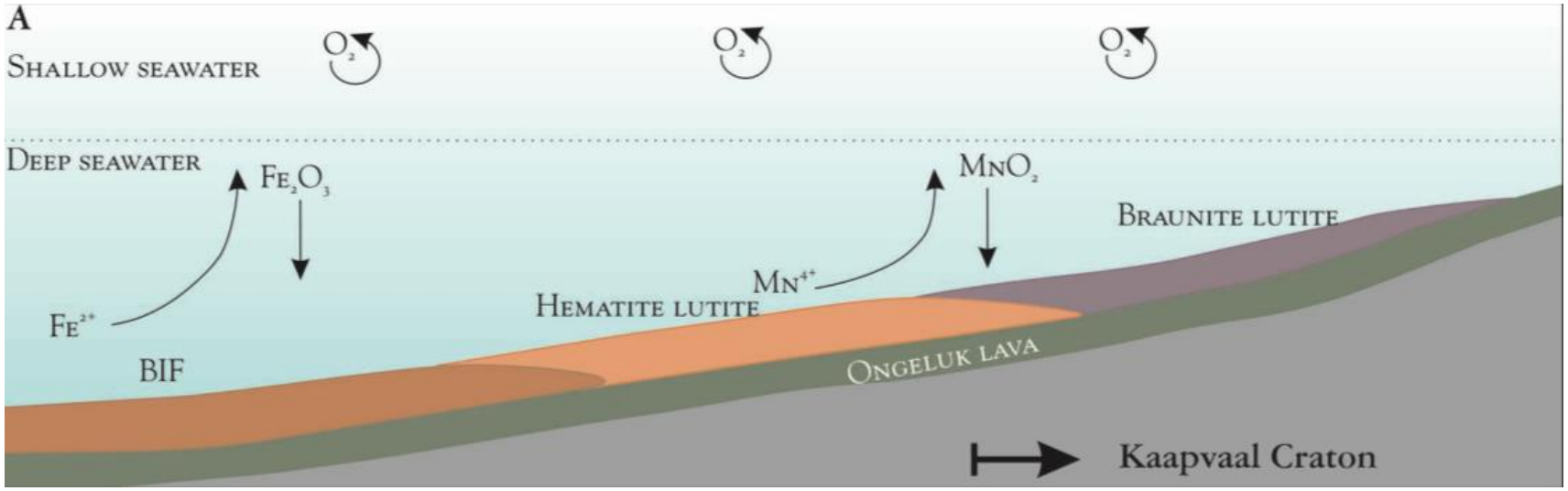
Source: http://www.aquilaresources.com.au



Source: http://www.nmmu.ac.za

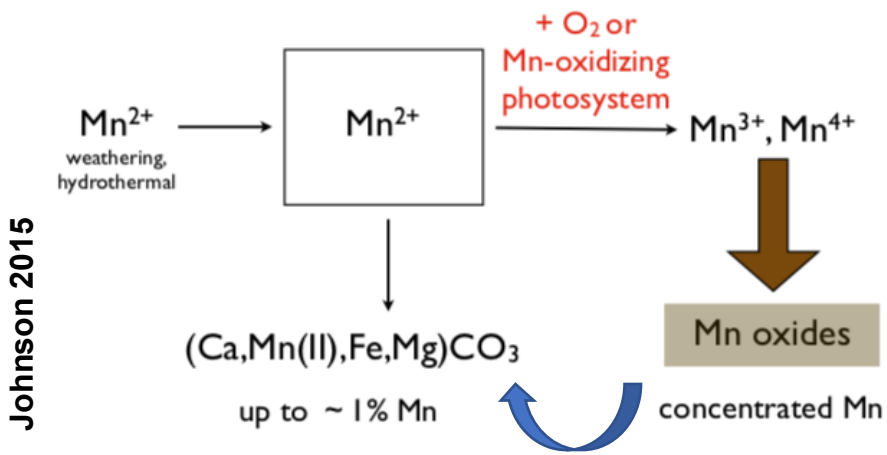
- Largest terrestrial Mn deposit
- Concentration 20-48 wt.% Mn, ore bodies <45m thick

Manganese formations
Depositional processes



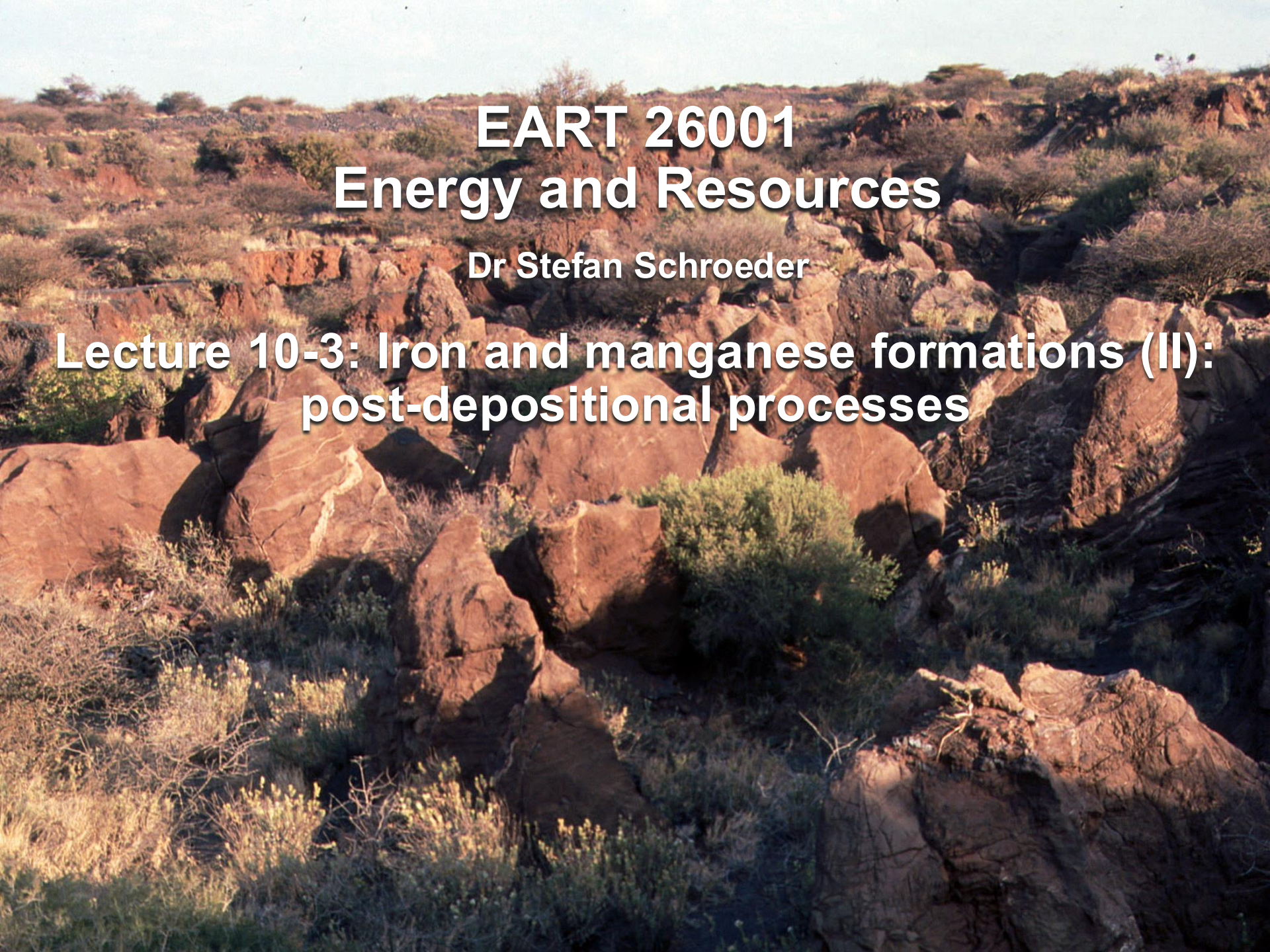
Schneiderhan 2006

$$Mn^{2+} + 0.25O_2 + 1.5H_2O \rightarrow Mn^{4+}OOH + 2H^+$$



Johnson 2015

$$2Mn^{4+}O_2 + CH_2O + 7H^+ \rightarrow 2Mn^{2+} + CO_3^{2-} + + 2H2O$$

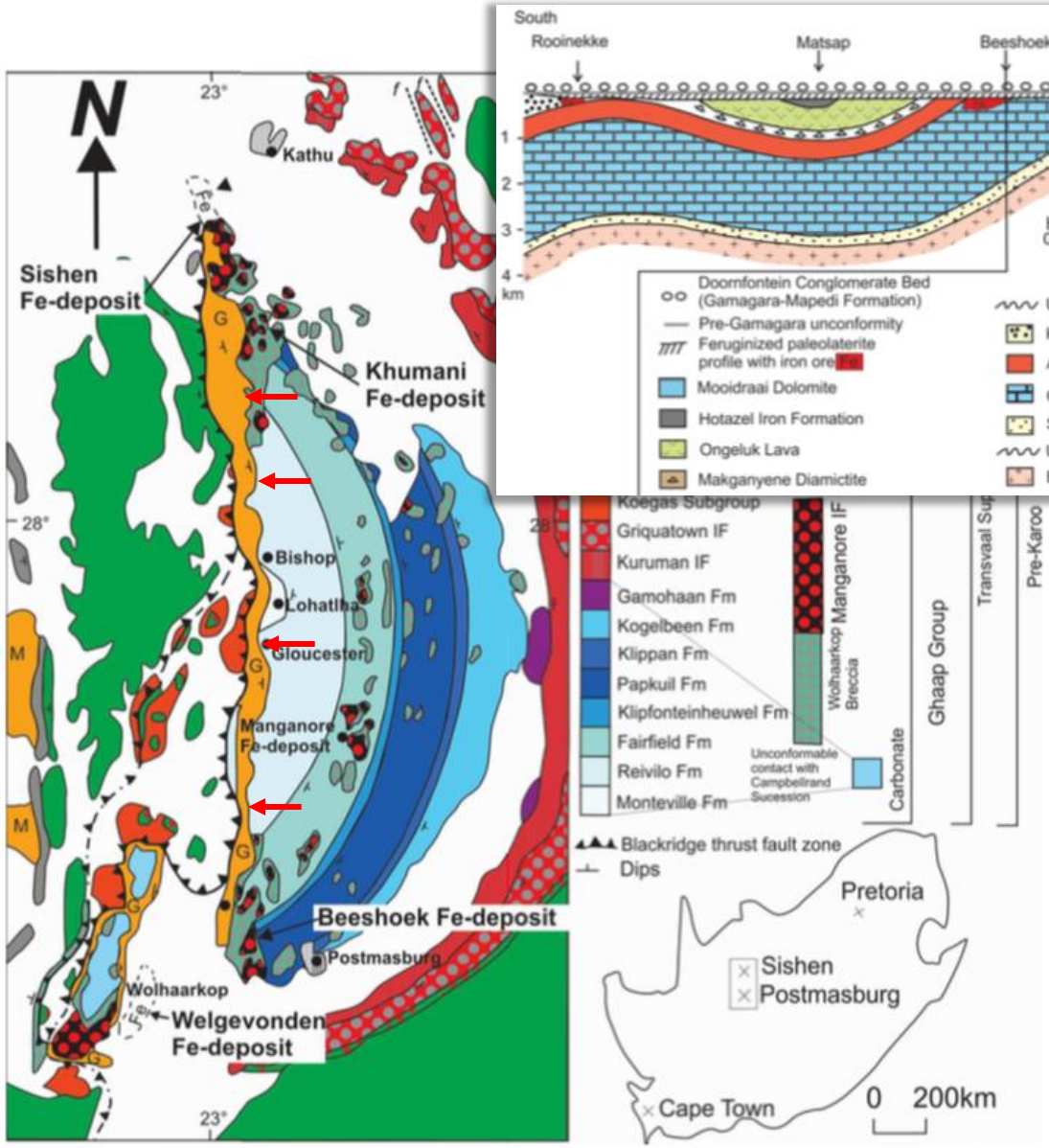


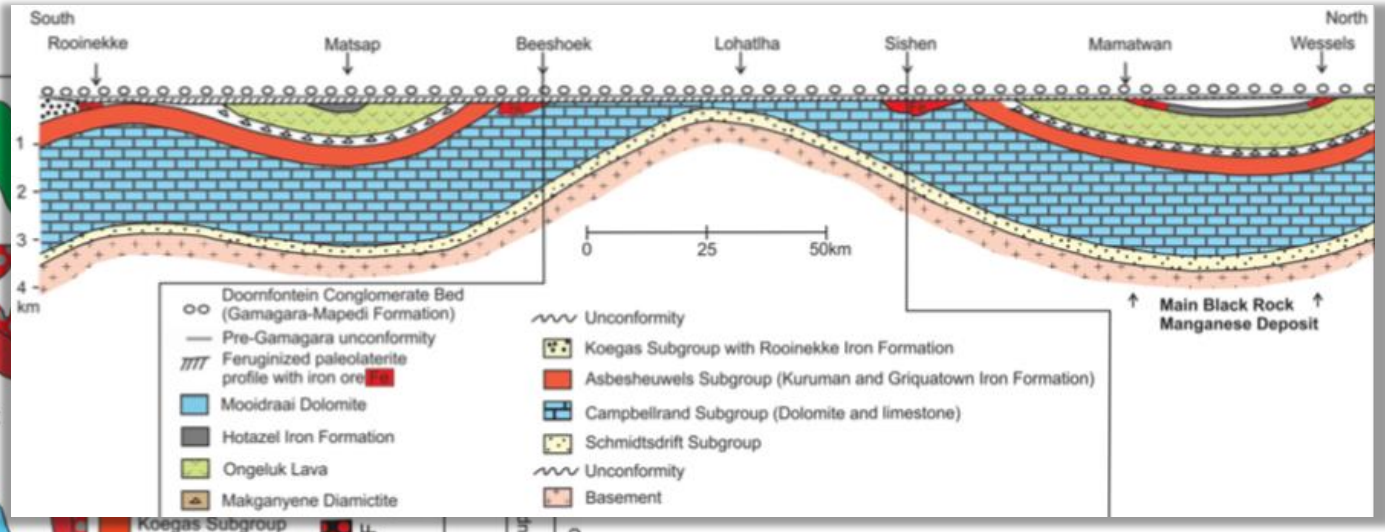
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Lecture 10-3: Iron and manganese formations (II): post-depositional processes

Iron and manganese formations
Supergene processes



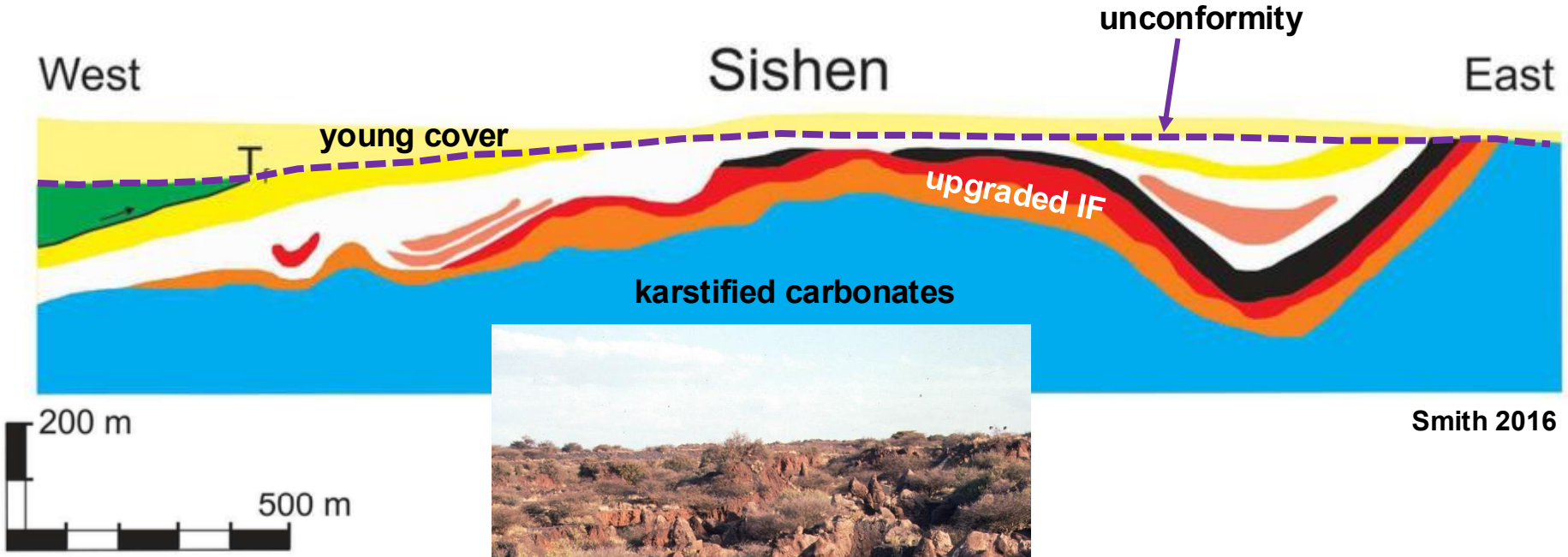


Smith 2016

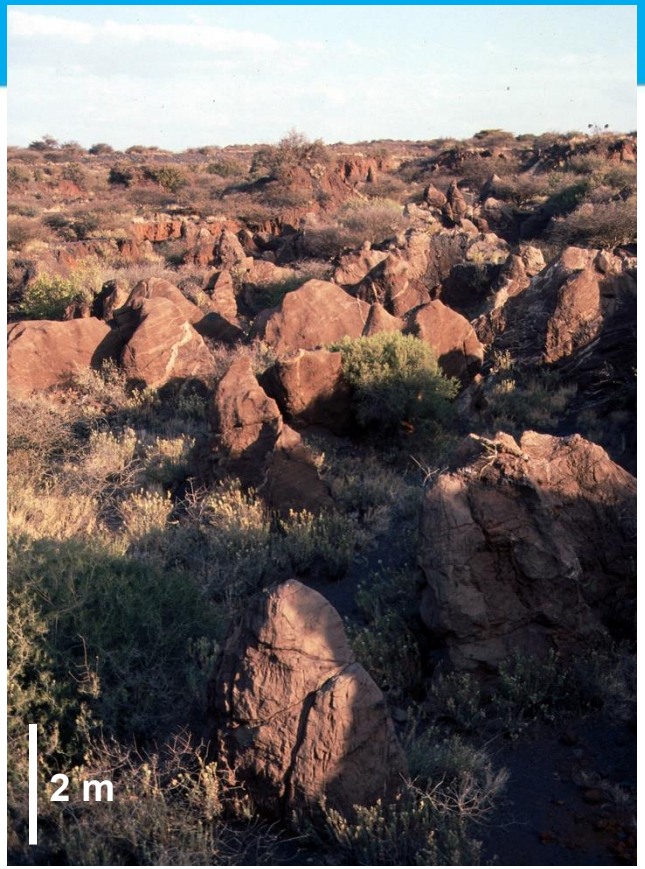
- Sedimentary processes of BIF and MnF deposition lead to deposits with max 40% Fe and 20% Mn, and high Si, P, Al contents
- Economically viable deposits need 30-65% Fe and Mn
- Supergene and hypogene fluid processes needed to upgrade

Smith 2016

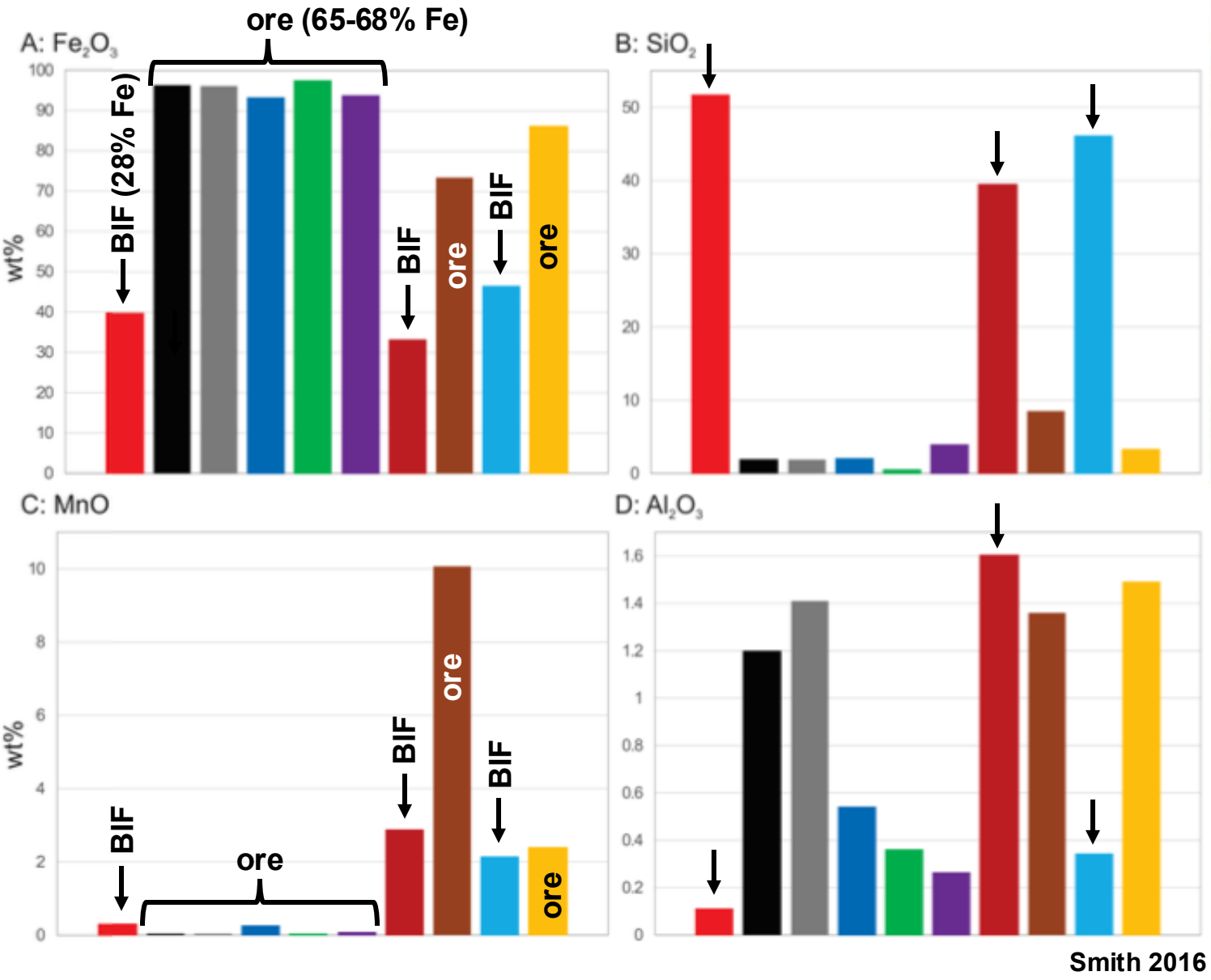
Iron and manganese formations
Supergene processes



Smith 2016



Iron and manganese formations
Supergene processes



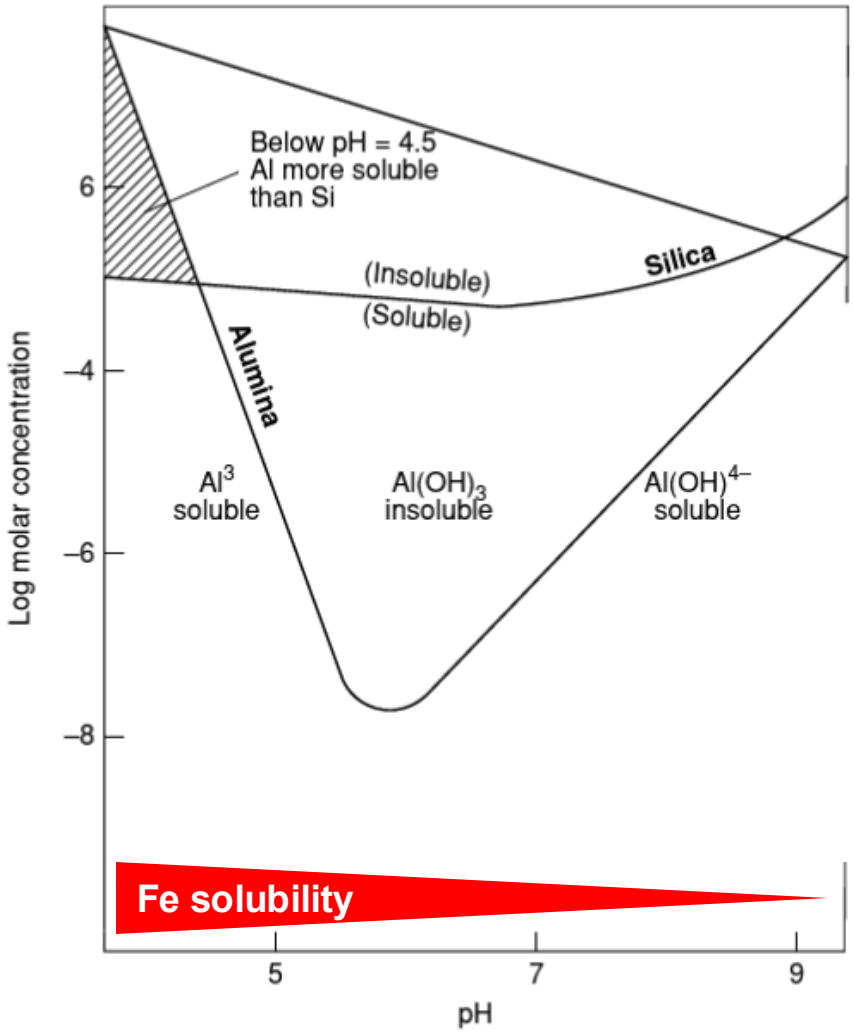


laminated ore



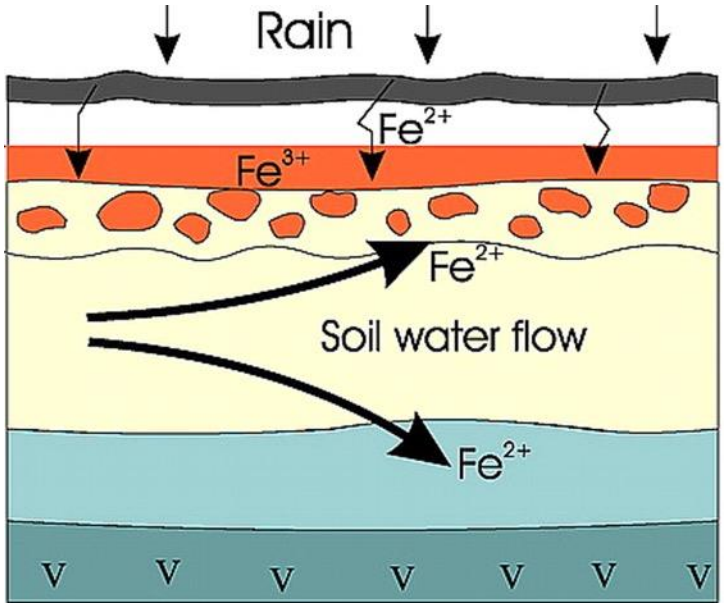
massive ore

Iron and manganese formations
Supergene processes



Robb 2005

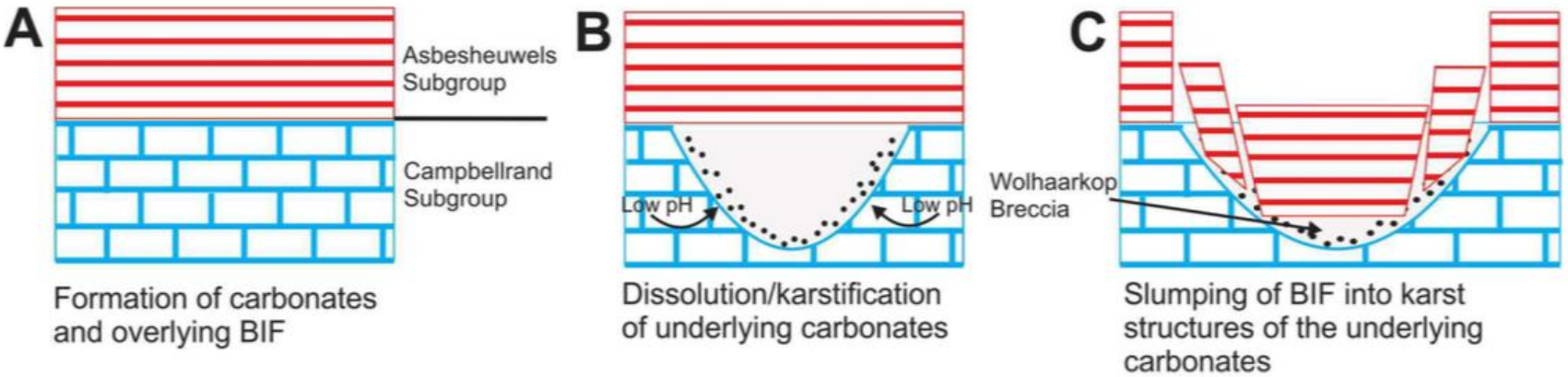
Laterite

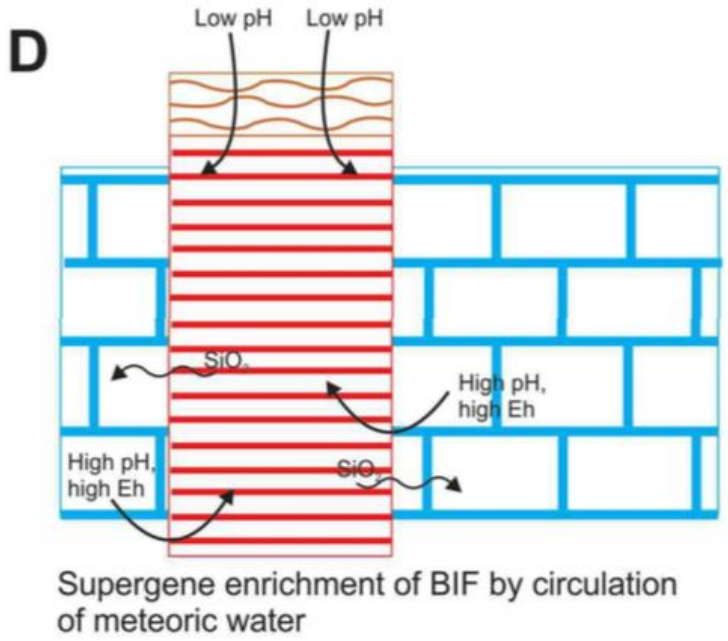


Beukes 2002



Iron and manganese formations
Supergene processes

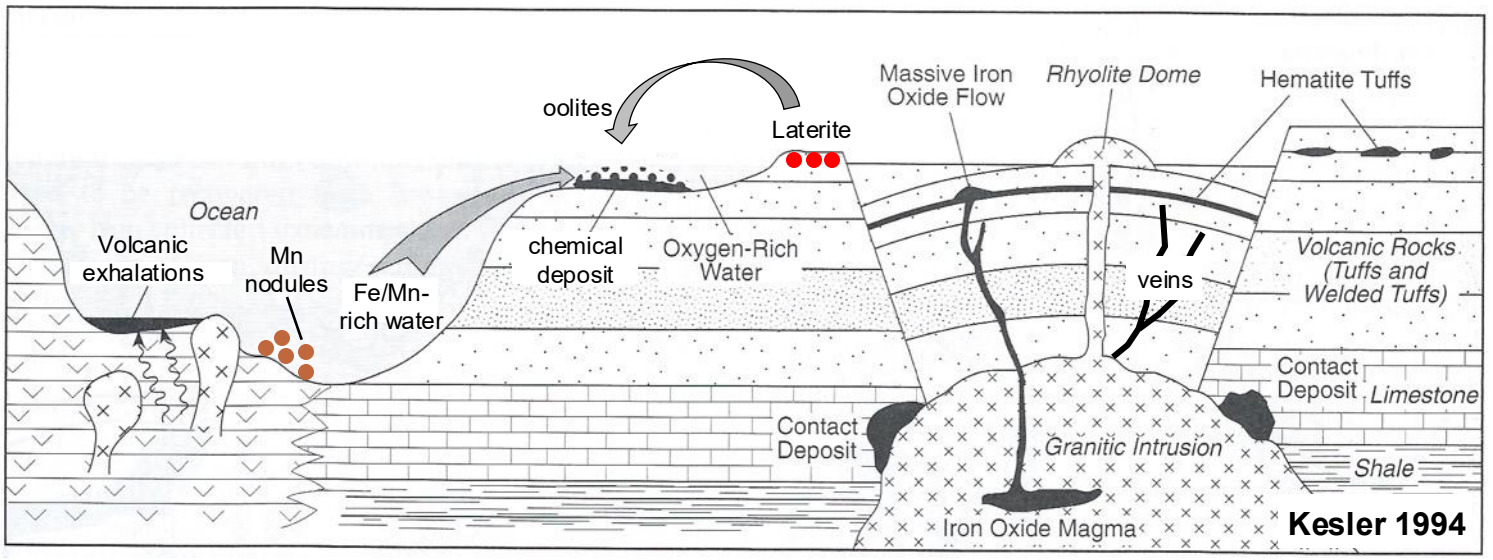




Smith 2016

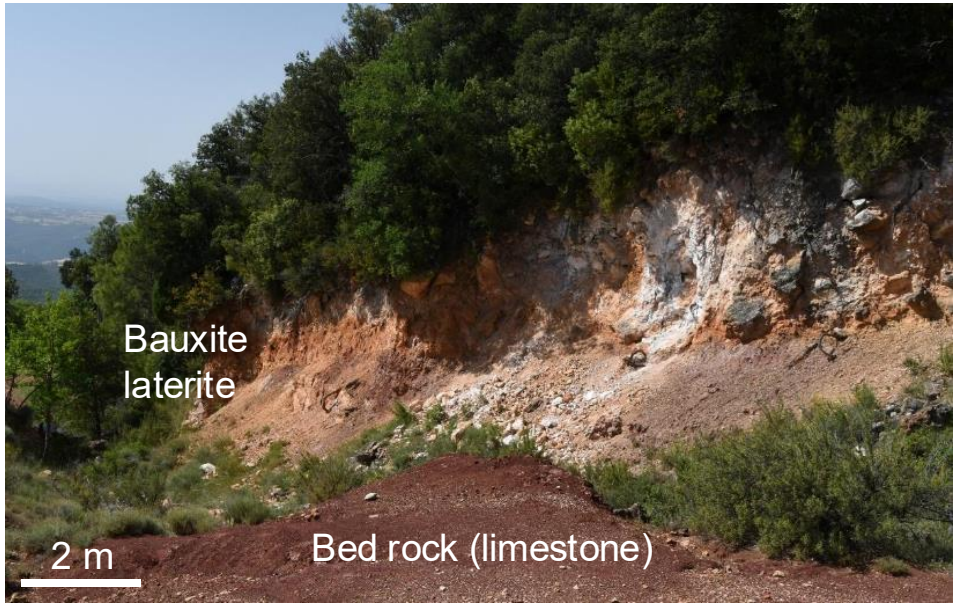
Iron and manganese formations
Summary

- Redox processes are essential in forming iron and manganese chemical sediments, and their diagenetic modification through biotic and abiotic processes.
- Banded iron and manganese formations of the Proterozoic are some of the largest iron and manganese deposits in the world, closely related to the unique environmental conditions of the first oxygenation of the atmosphere.
- Deposit types:
  - Chemical marine, incl Mn nodules and Fe oolites
  - Laterite
  - Marine volcanic exhalations
  - Hydrothermal veins
  - Contact metamorphic (skarn)
  - Orthomagmatic



Surficial resource deposits
Aluminium in bauxite





© S Covey-Crump, Manchester

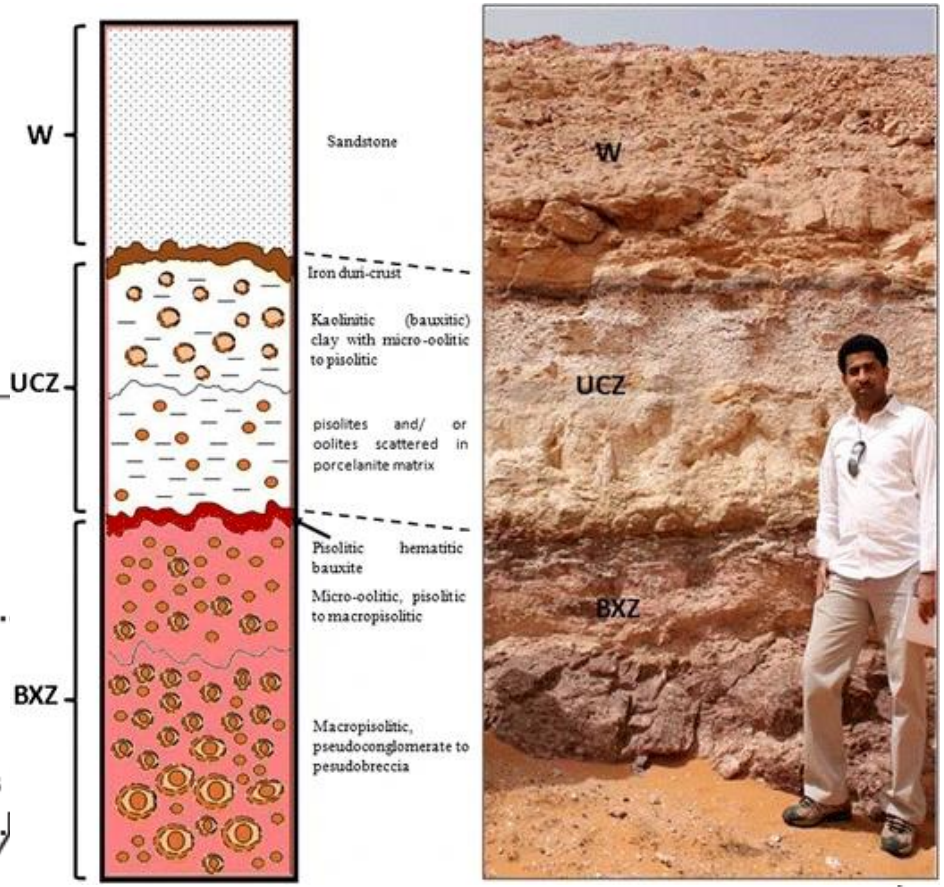
- Very abundant metal
- Low density and high strength/weight ratio
- Good heat conductivity
- Good corrosion resistance
- Used as lightweight construction material (aerospace) and for insulation
- Main resource: Bauxite



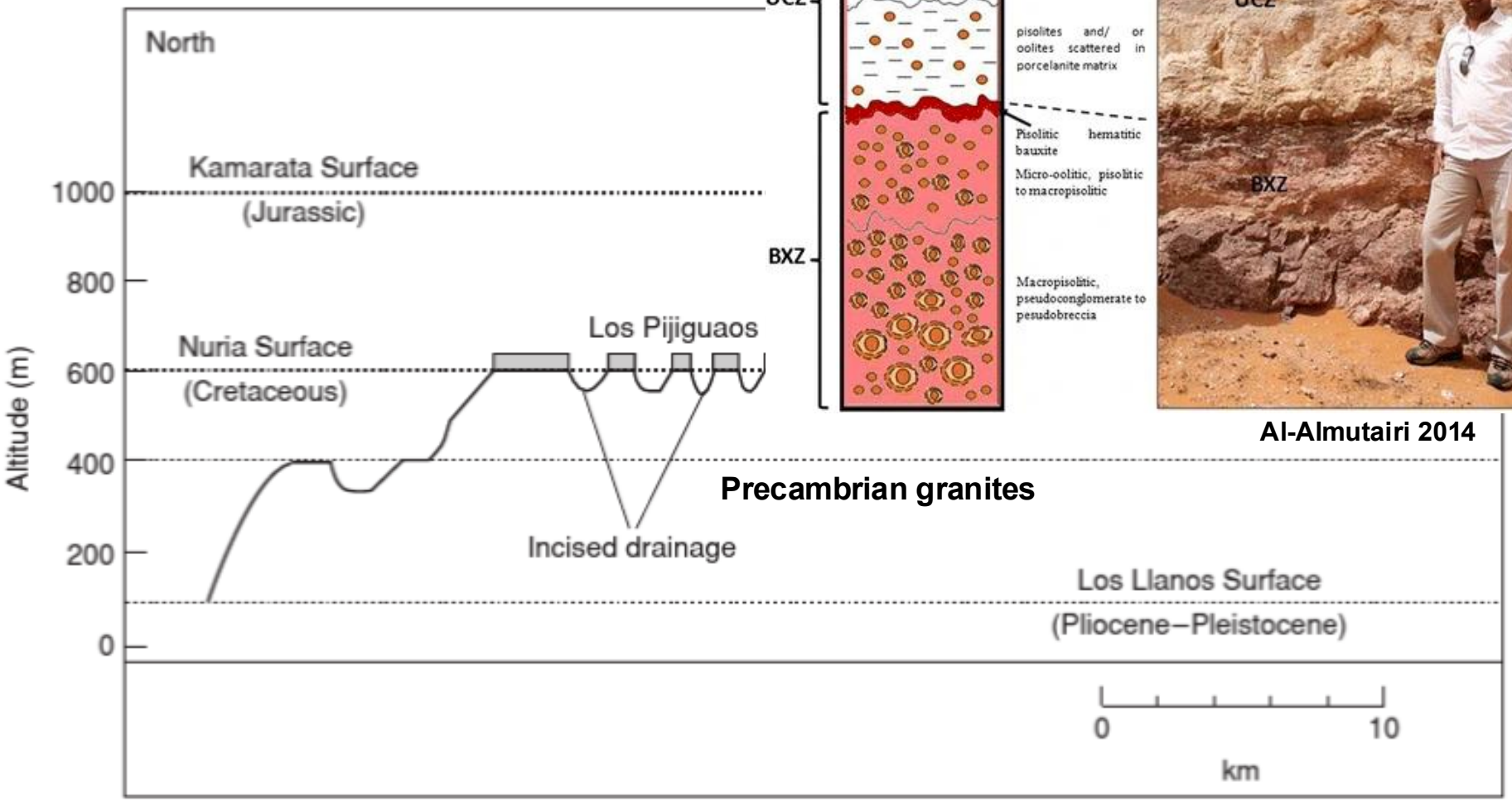
Source: USGS

Surficial resource deposits
Aluminium in bauxite

- (Long-term) weathering and karstification of various lithologies

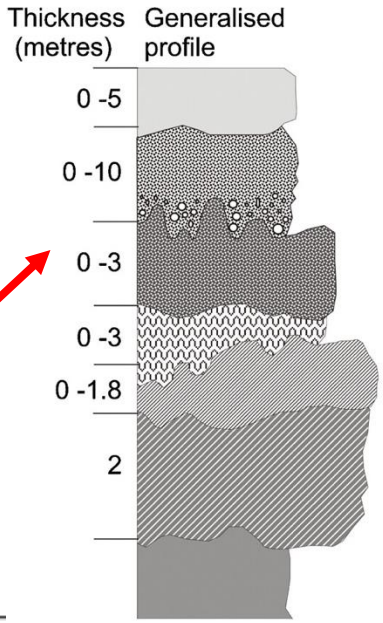


Al-Almutairi 2014



Robb 2005

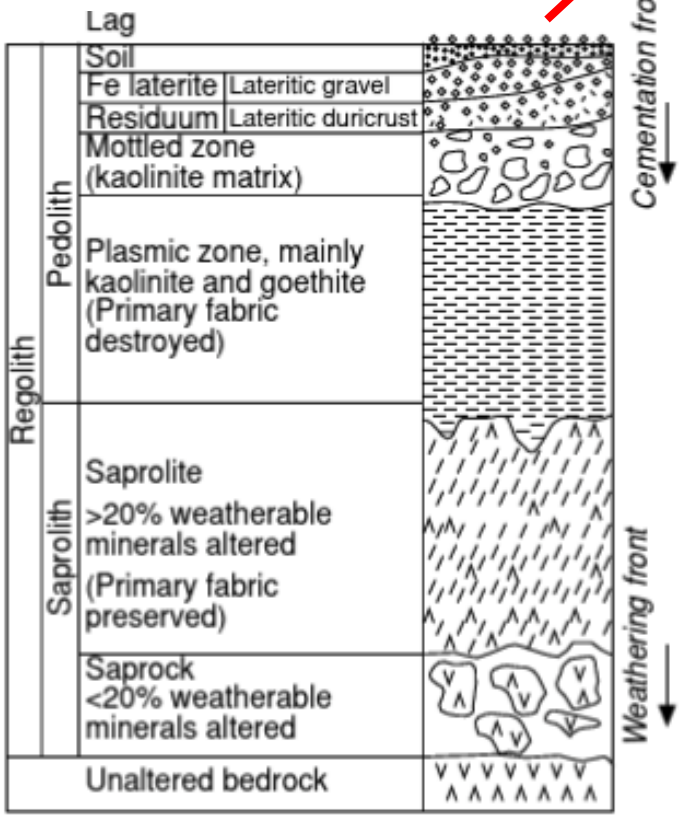
Surficial resource deposits
Aluminium in bauxite



Explanation		Characteristics Al_2O_3	SiO_2
Soil			
Loose pisolitic		46%	6%
Upper nodules	(Fig. 5b)		
Cemented pisolitic	(Fig. 5c)	49%	4%
Tubular	(Fig. 5d)	52%	2%
Lower nodules		50%	2%
Nodular ironstone		25%	35%
Mottled clay zone			

Bauxite

Abzalov 2014

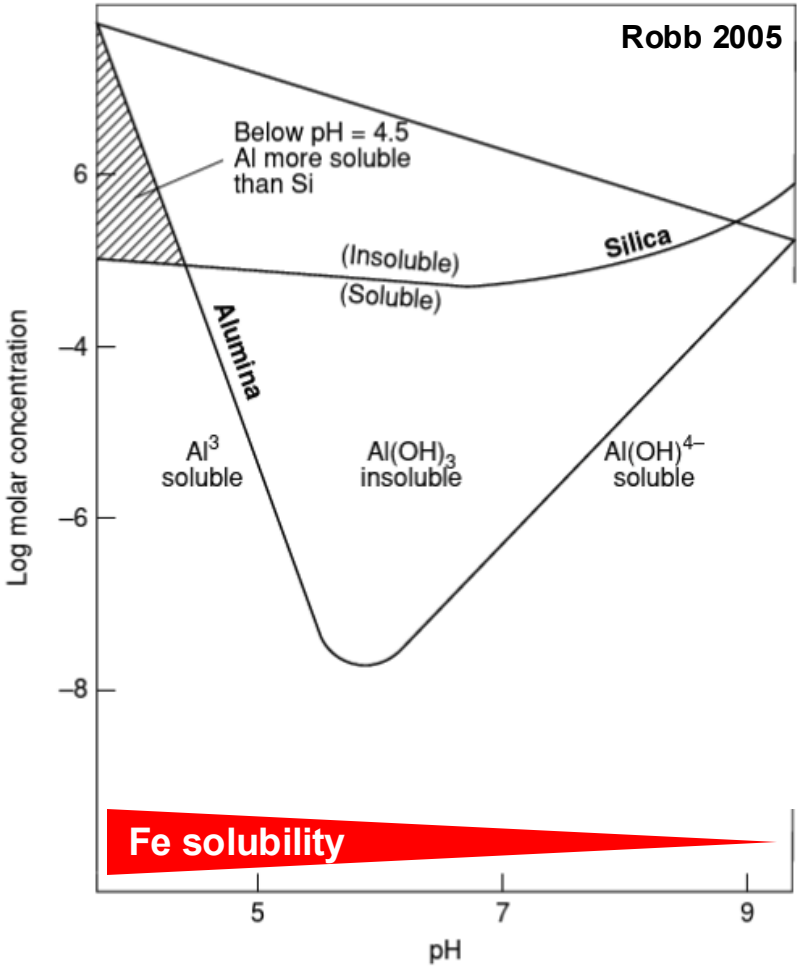


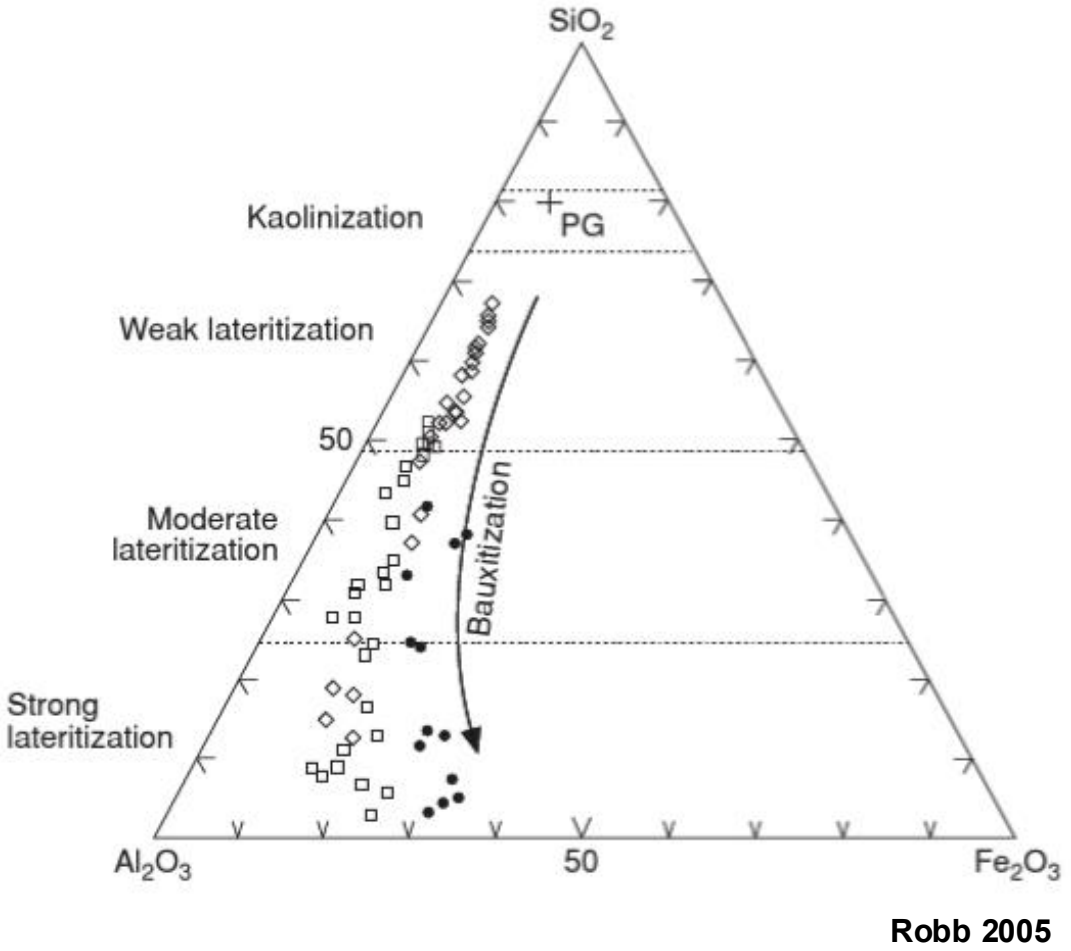
Host minerals
Released in the mottled and ferruginous zones
Aluminosilicates (muscovite, kaolinite) iron oxides; gold
Released in upper saprolite
Aluminosilicates (muscovite)
Ferromagnesians (chlorite, talc, amphibole)
Smectite clays
Released in the lower saprolite
Aluminosilicates
Ferromagnesians (pyroxene, olivine amphiboles, chlorite, biotite)
Released at weathering front
Sulfides
Carbonates

Robb 2005

- **Regolith soil formation**
- **Progressive removal of Si and relative enrichment of Al**
- **Humid climate: laterite and bauxite formation (Fe dissolution)**
- **Semi-arid climate: Fe more stable – ferricrete (cf supergene BIF enrichment)**

Surficial resource deposits
Aluminium in bauxite



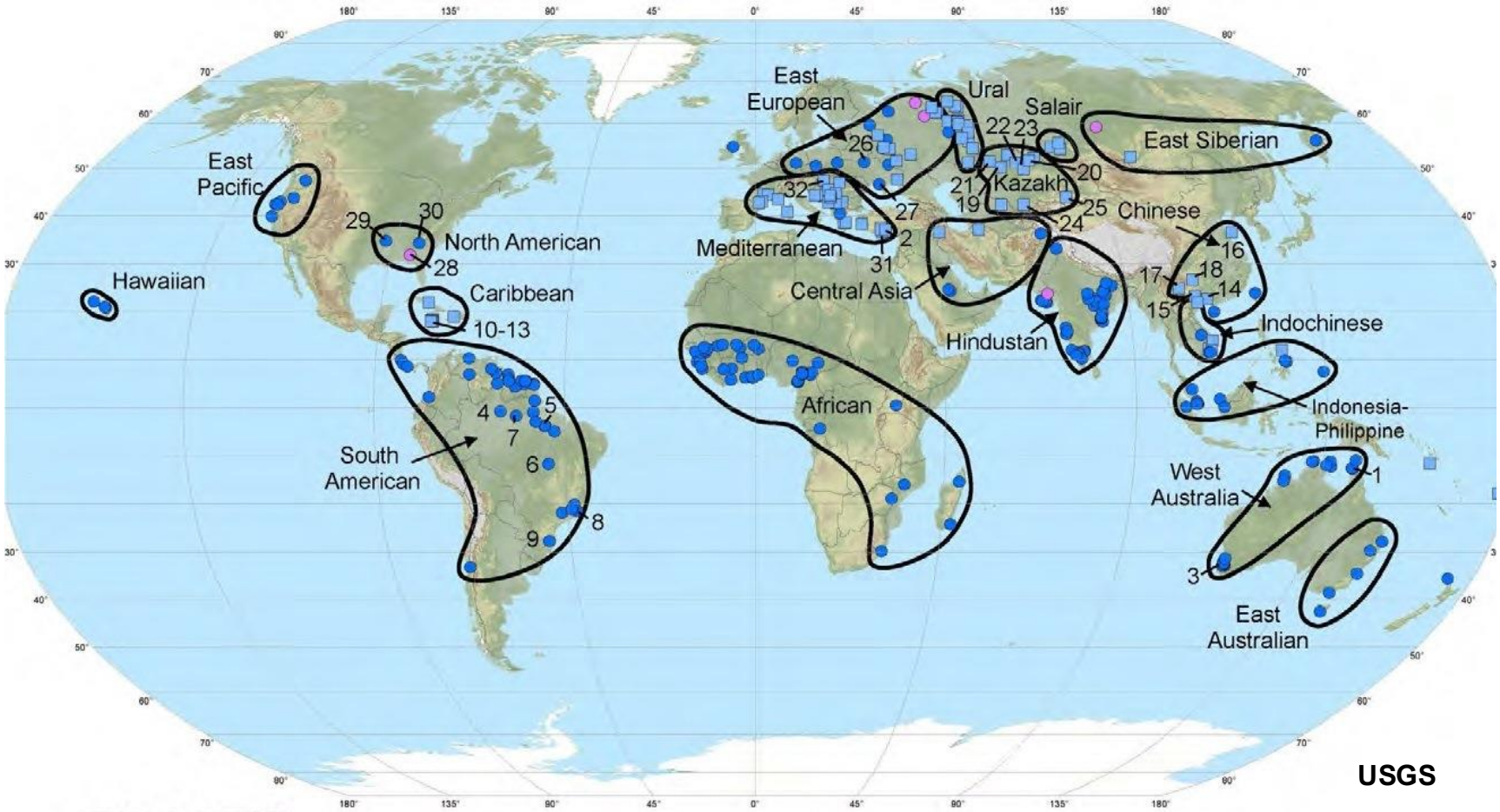


Robb 2005

feldspar $KAlSi_3O_8$ + H_2O (hydrolysis) – (loss of Si) -> kaolinite $Al_2Si_2O_5(OH)_4$ – (loss of Si) -> gibbsite $Al(OH_3)$

Surficial resource deposits
Aluminium in bauxite

Global bauxite distribution

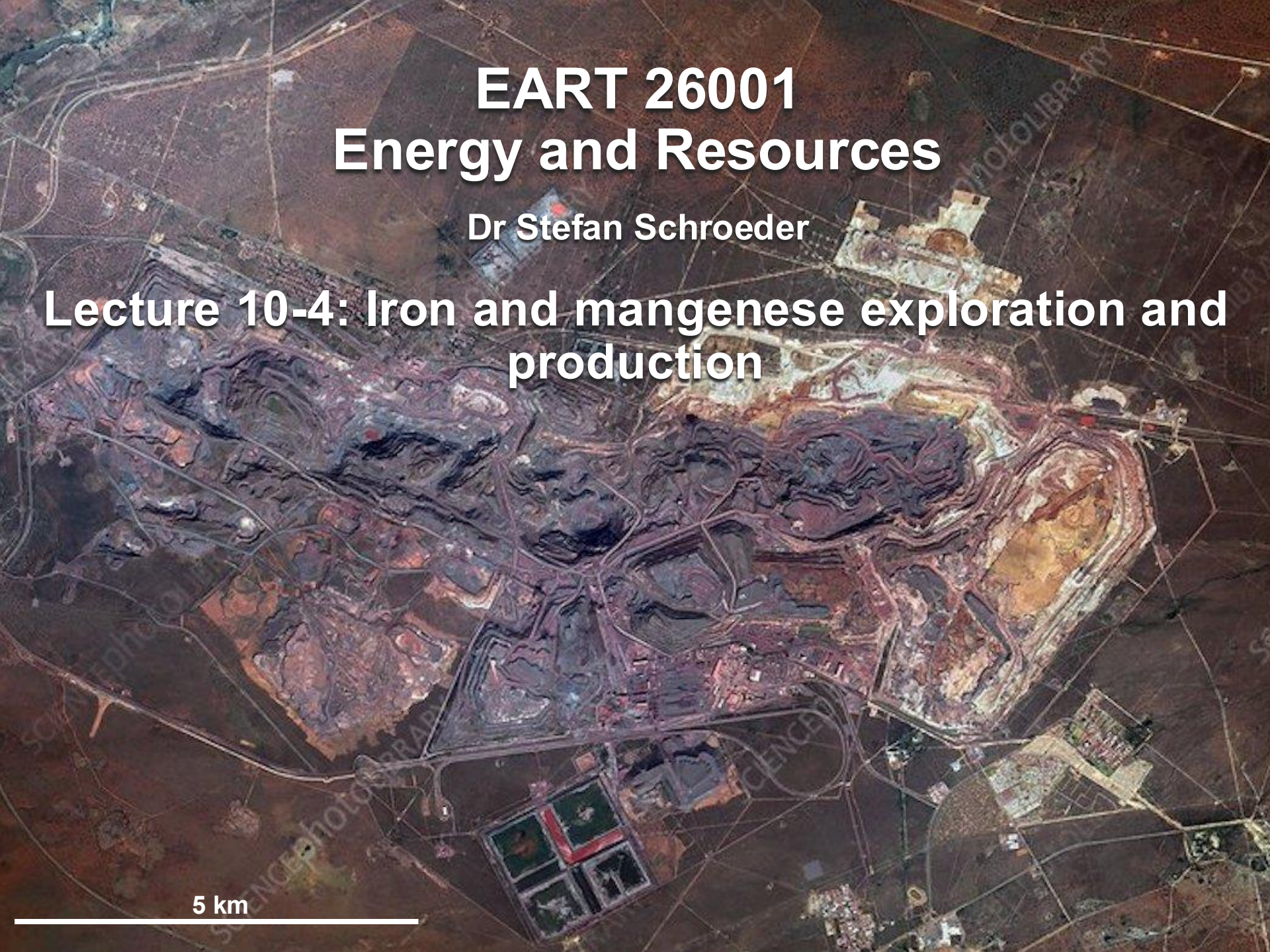


- **Bauxite and its processing product red mud are equally important secondary sources of REE (adsorbed to Fe oxides)**
- **Other residual laterite commodities: Ni, REE, Au**

EART 26001
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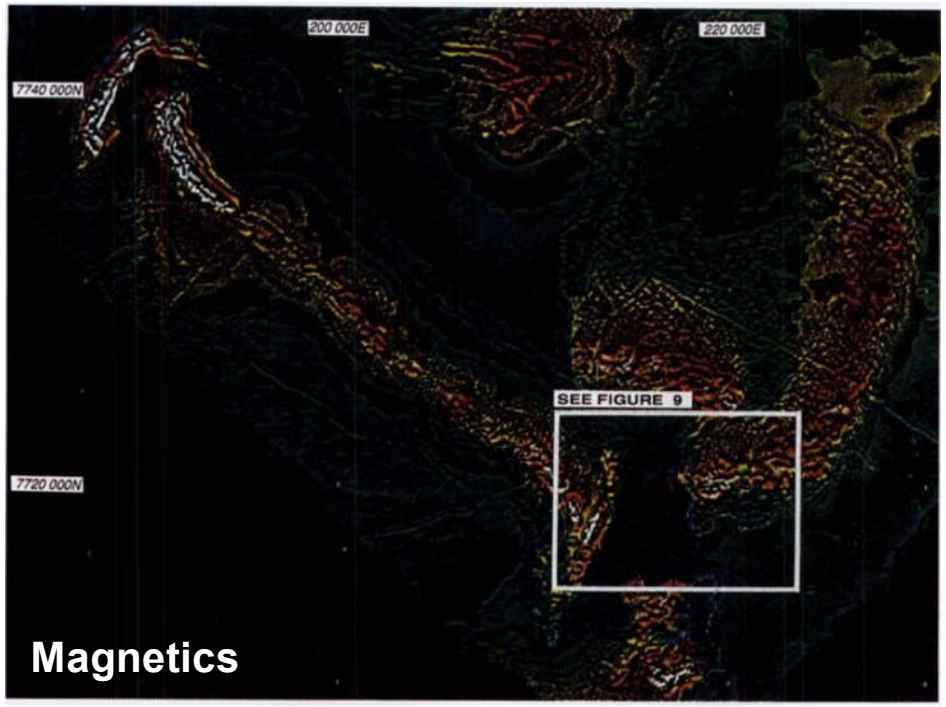
Dr Stefan Schroeder

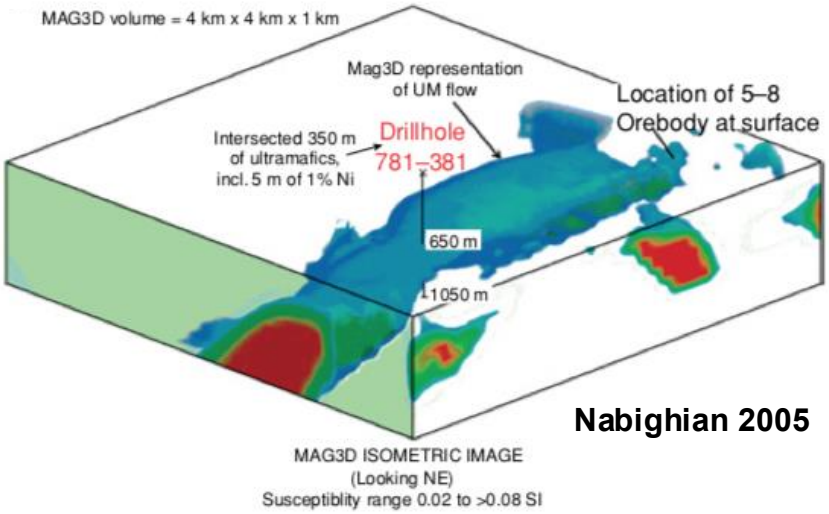
Lecture 10-4: Iron and mangenese exploration and production



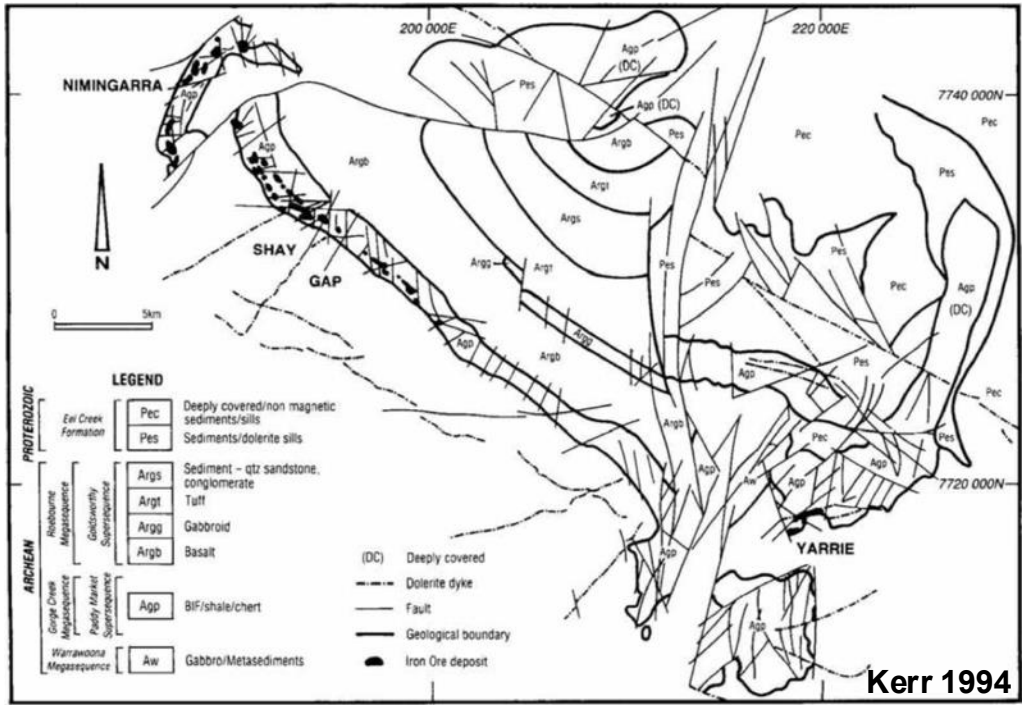
Exploration
Geophysical techniques

- Magnetic and magnetic susceptibility surveys exploit the elevated magnetization of magnetite
- Gravity surveys depend on the elevated density of iron and manganese minerals relative to country rocks





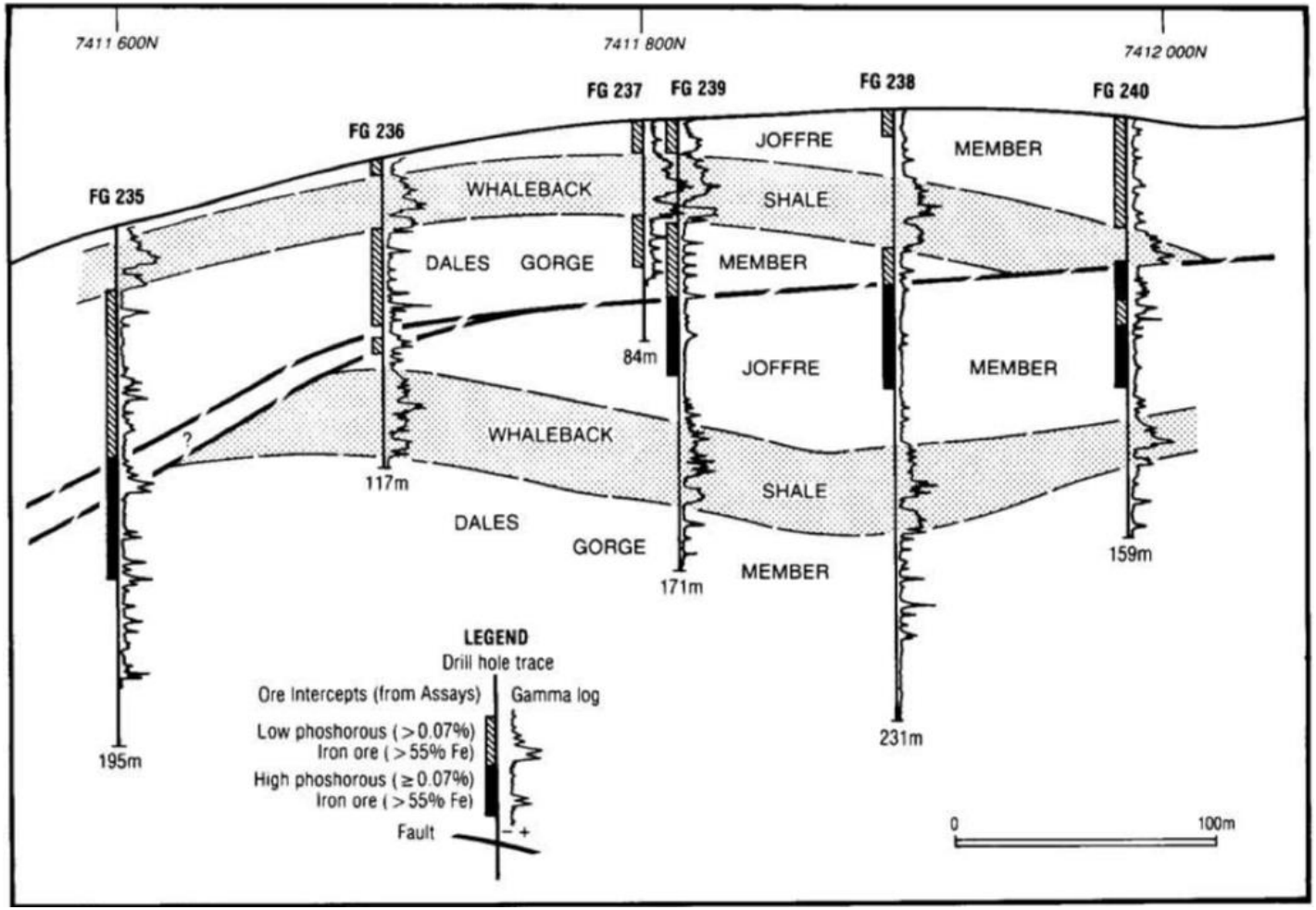
Nabighian 2005



Kerr 1994

Exploration
Geophysical techniques

Logging

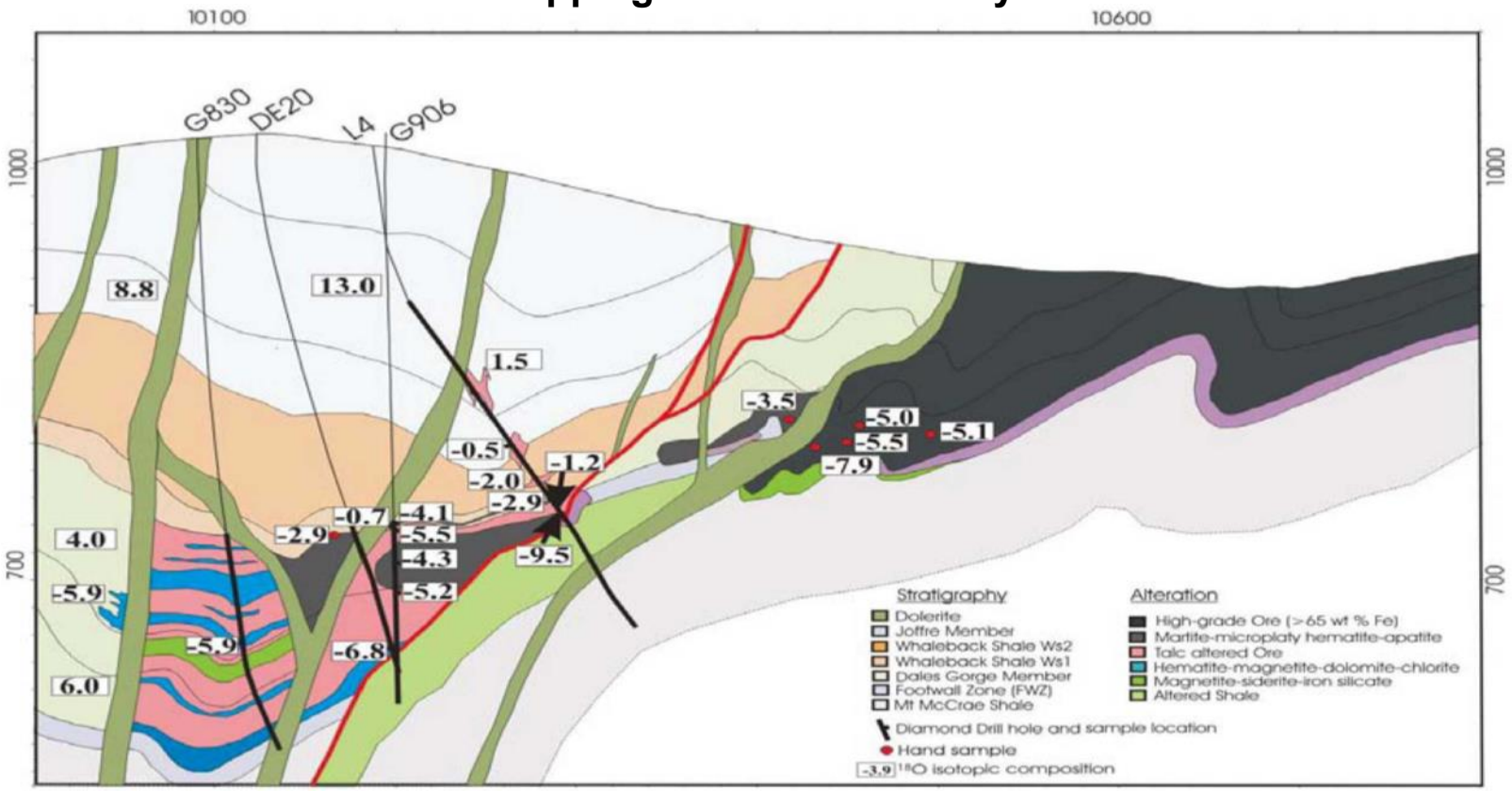


Kerr 1994

- Gamma-ray logging in drill core uses the natural radioactivity

Exploration
Mapping and geochemistry

Mapping and Geochemistry

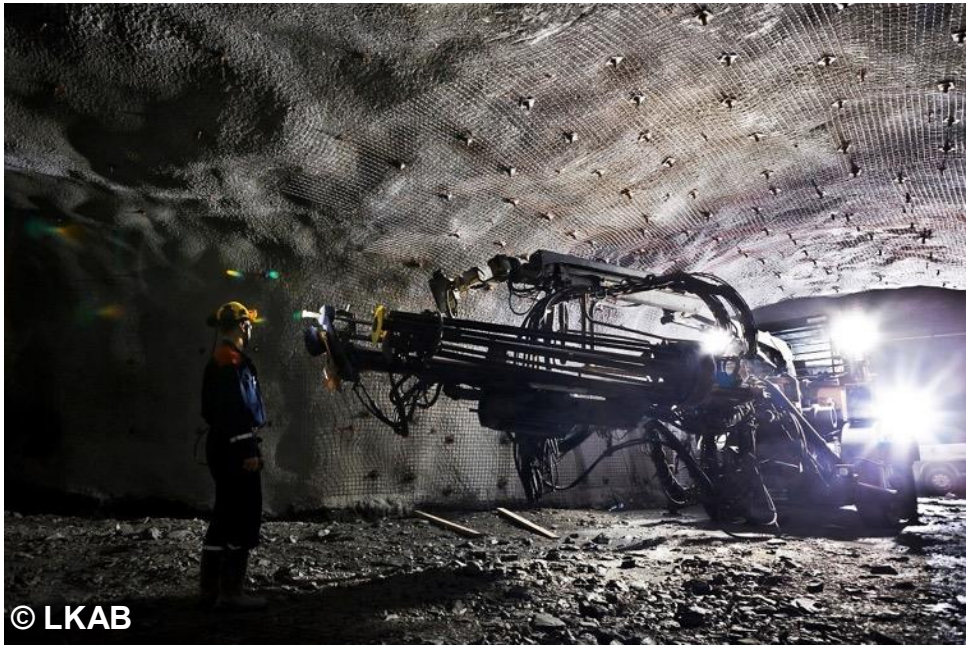


Hagemann 2007

- **Geochemistry:**
  - **Mapping ore grades**
  - **Constraining genetic conditions (p-T-X) and alteration zones**
- **Mapping is essential for structural and sedimentary context**

Production
Mining

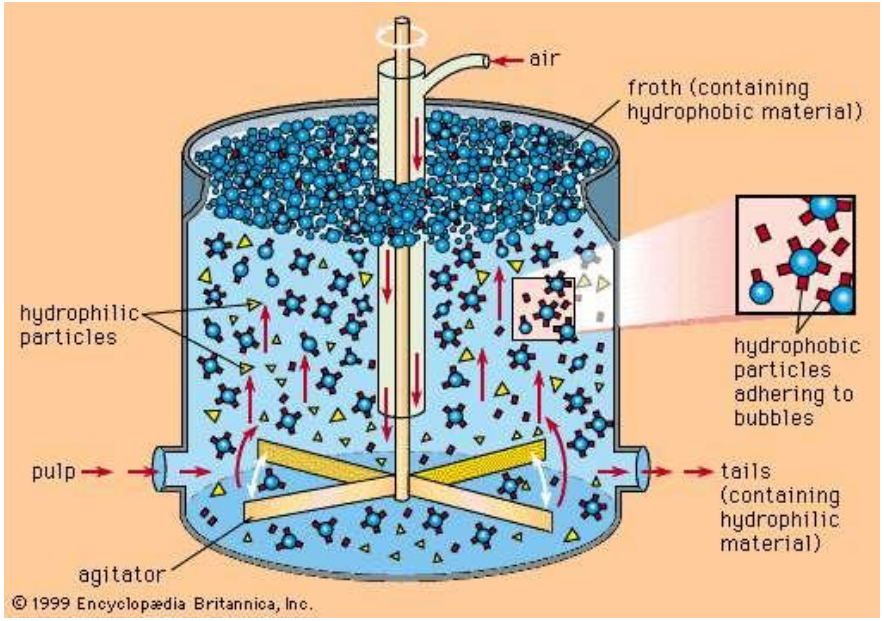




- Soft ore: dominance of open pit mining
- Hard ores/vein ores (e.g. Kiruna): underground mining

Production
Beneficiation

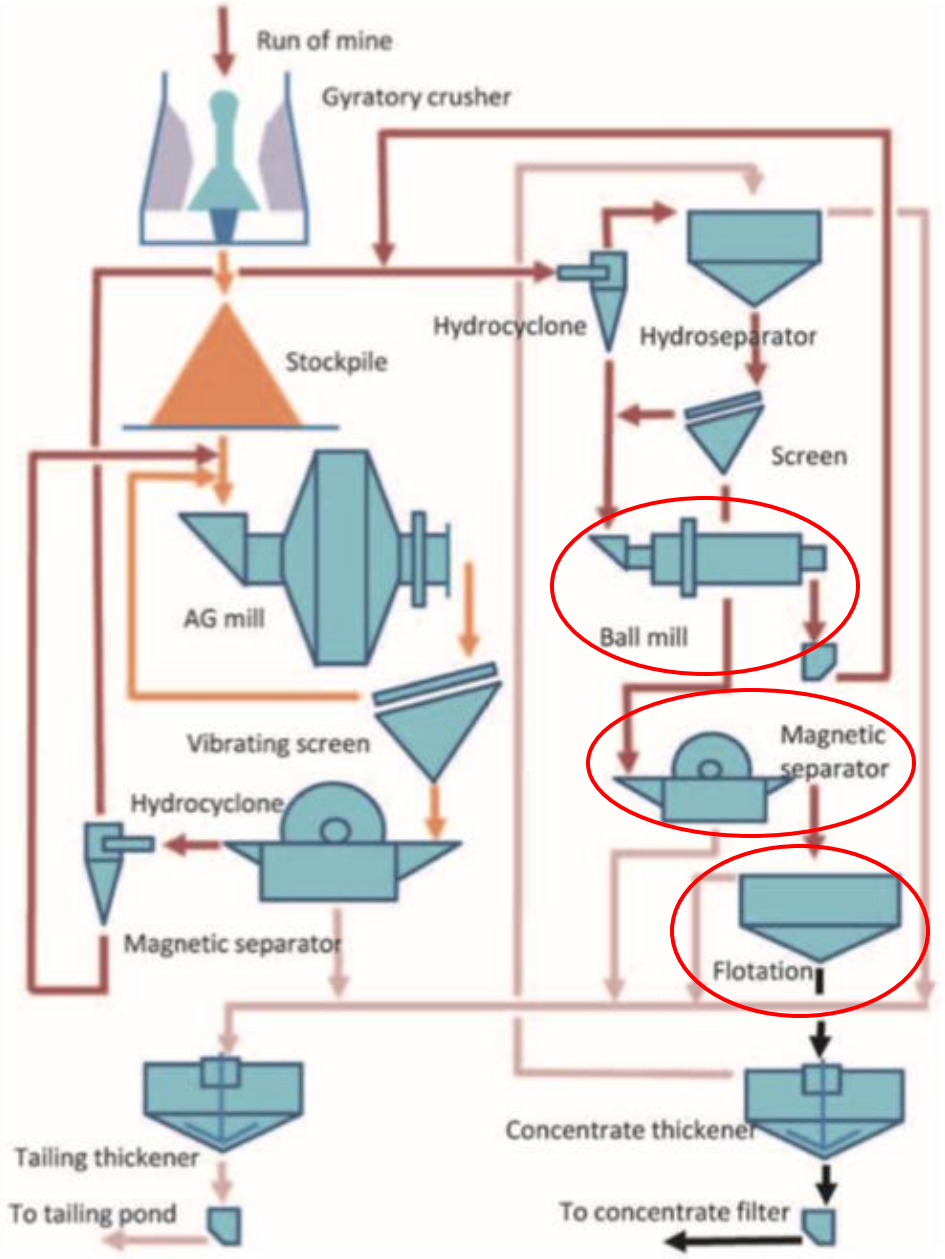
Flotation



© Encyclopedia Britannica

- Concentrate ore (>55-65% Fe) is pelletized or sintered for shipping to steelworks

Processing workflow



Nomura 2015

Environmental aspects
Mining activity

Australians

Rio Tinto blasts 46,000-year-old Aboriginal site to expand iron ore mine

Mining company was given permission to blast Juukan Gorge cave, which provided a 4,000-year-old genetic link to present-day traditional owners

Calla Wahlquist

@callapilla

Tue 26 May 2020
11.59 BST

35k

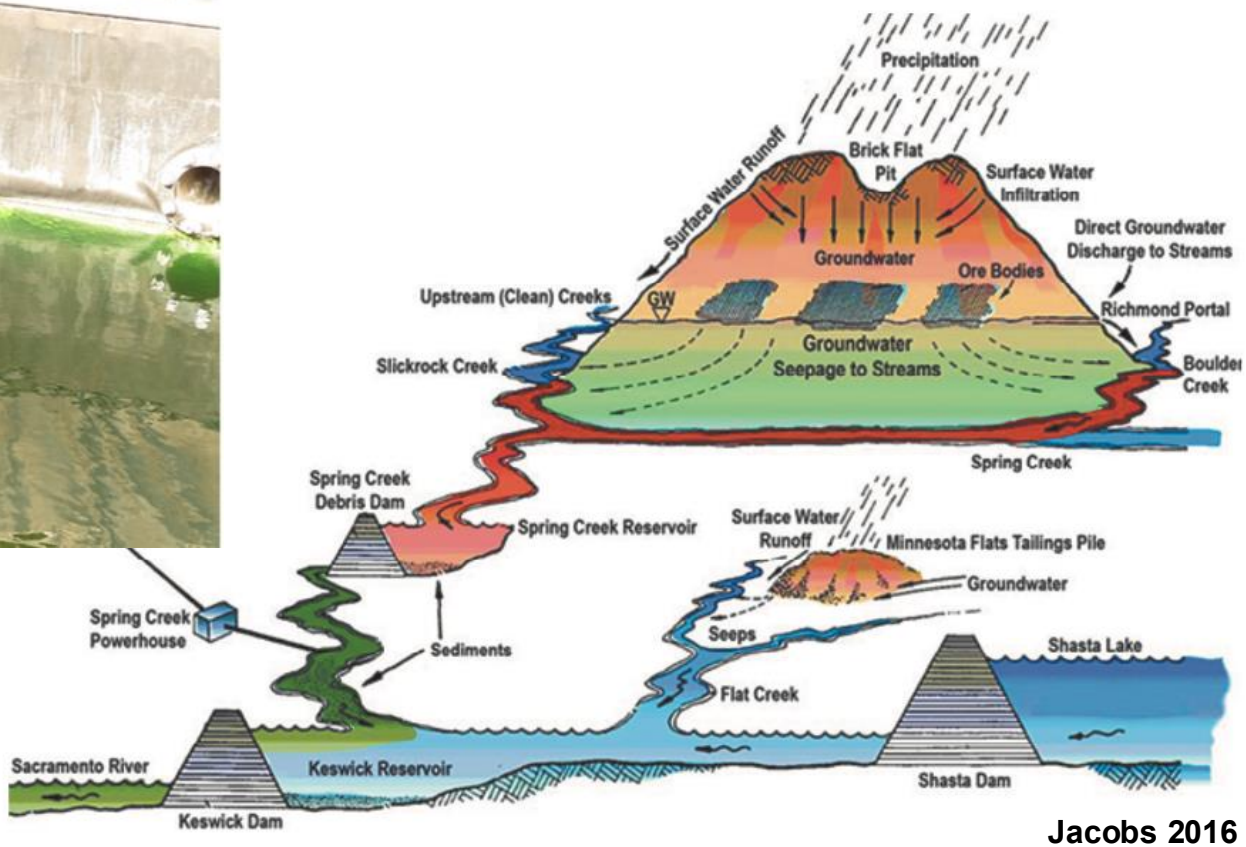


© Guardian

Environmental aspects
Acid mine drainage

- For sulfidic (iron) ores
- $4FeS_2 + 15O_2 + 8H_2O \rightarrow 2Fe_2O_3 + 8SO_4^{2-} + 16H^+$

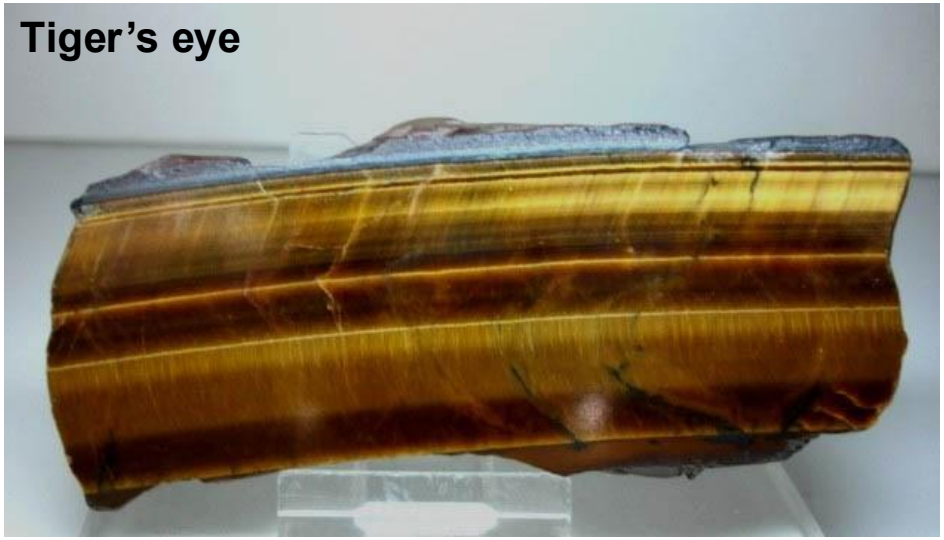




Jacobs 2016

Environmental aspects
Asbestos

- Fibrous mineral varieties of
  - Serpentine: chrysotile $Mg_6(OH)_8(Si_4O_{10})$
  - Amphibole: riebeckite (crocidolite) $Na_2(Mg,Fe^{2+})_3Fe_2^{3+}(OH)_2(Si_8O_{22})$
    anthophyllite $(Mg,Fe^{2+})_7(OH)_2(Si_8O_{22})$
    and others
- Common byproduct of metamorphism of banded iron formations
- Mined till 1980s for heat resistant properties
- Carcinogen, silicosis



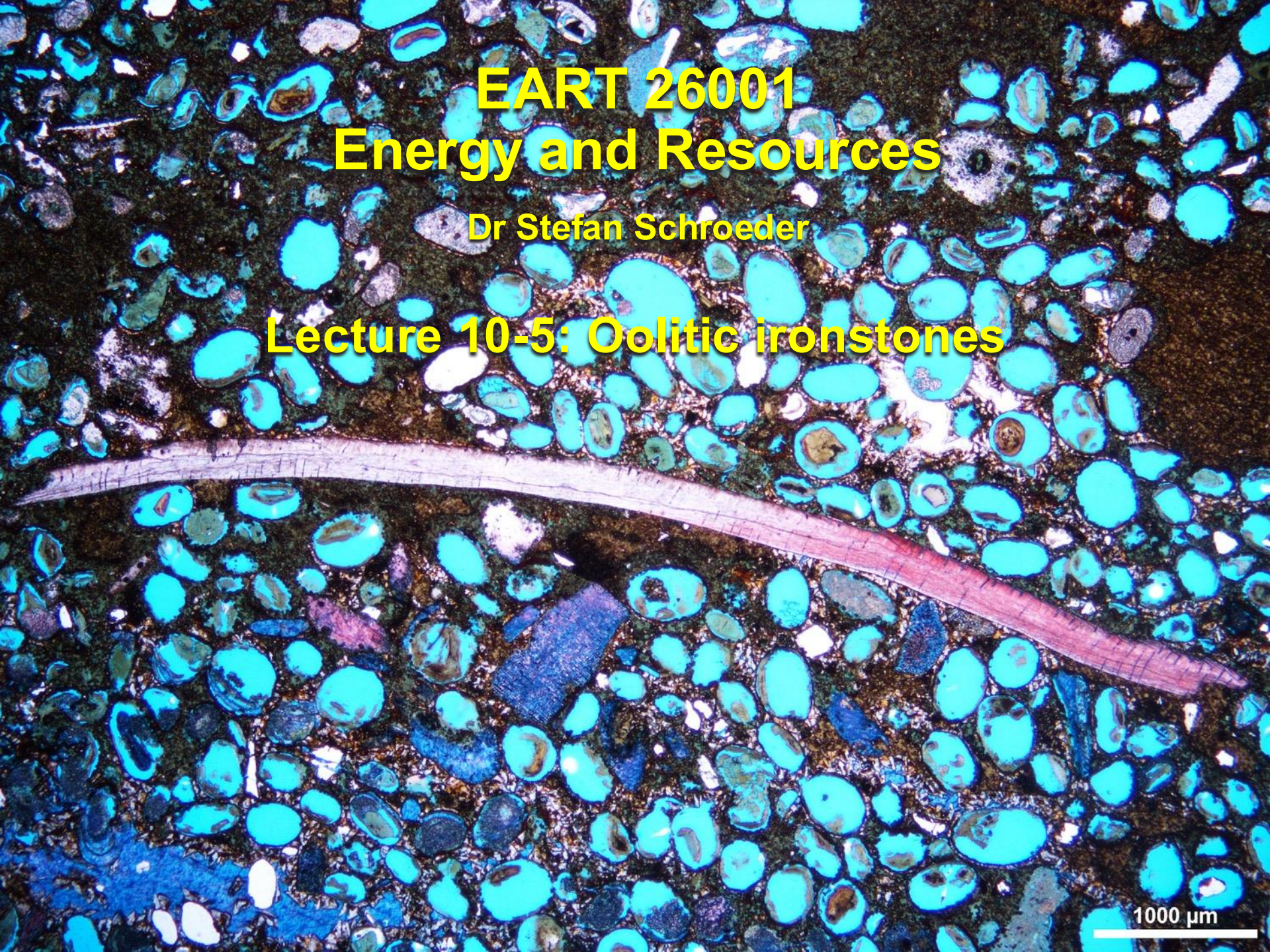
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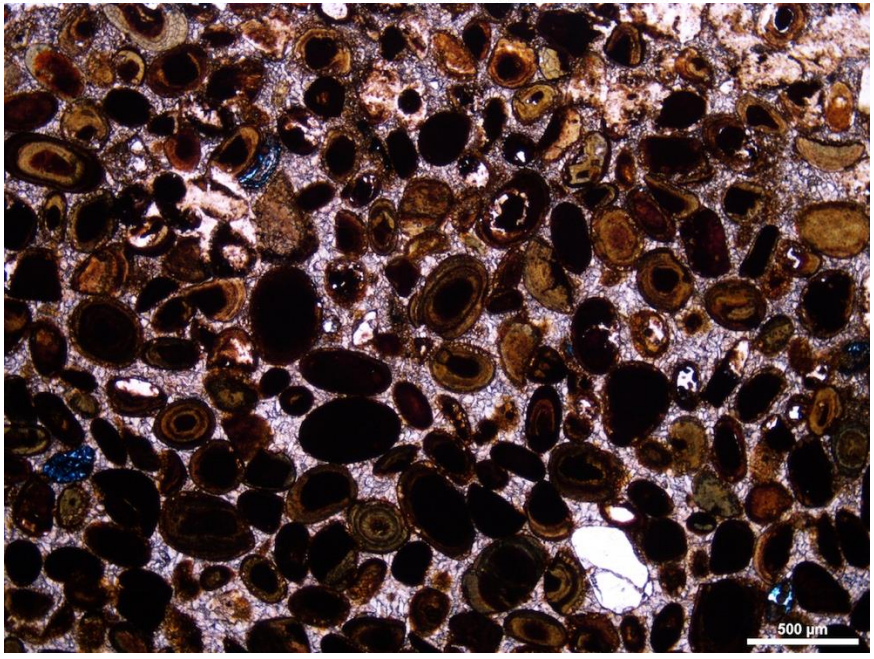
Iron and manganese deposits
Summary week 10

- Iron and manganese are mostly used in the steel and construction industry; further applications are in battery manufacturing and chemical industry.
- They are relatively abundant in the crust, and enrichment factors are low compared to other metals.
- Their redox chemistry (soluble at low redox state, insoluble at high redox state), and association with mafic magmas (for Fe) largely determine deposits.
- Iron and manganese occur in a variety of deposit types:
  - Magmatic and volcanic exhalations, metamorphic
  - Hydrothermal
  - Marine chemical sediments
  - Laterites
- Major iron and manganese formations in the Proterozoic are also key for scientific understanding of the early atmosphere.


EART 26001
Energy and Resources
Dr Stefan Schroeder
Lecture 10-5: Oolitic ironstones
1000 µm

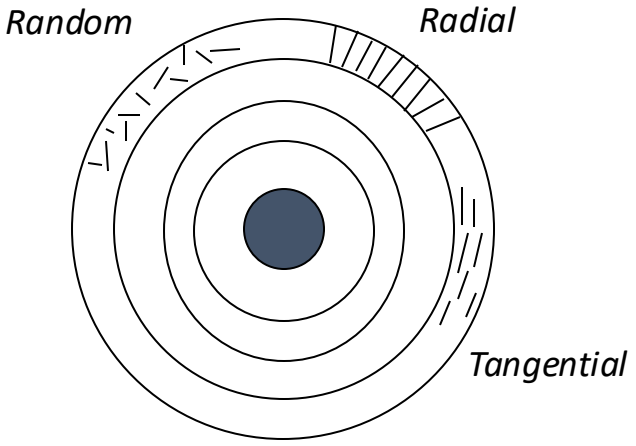
Oolitic ironstones
Petrography





- **Bioclastic sandstones with Fe-rich ooids**
- **Ooids consist of:**
  - Siderite ($FeCO_3$)
  - Chamosite ($Fe(Si,Al)_2O_5(OH)_4$)
  - Berthierine ($Fe^{2+}Fe^{3+}Al)_3(Si,Al)_2O_5(OH)_4$

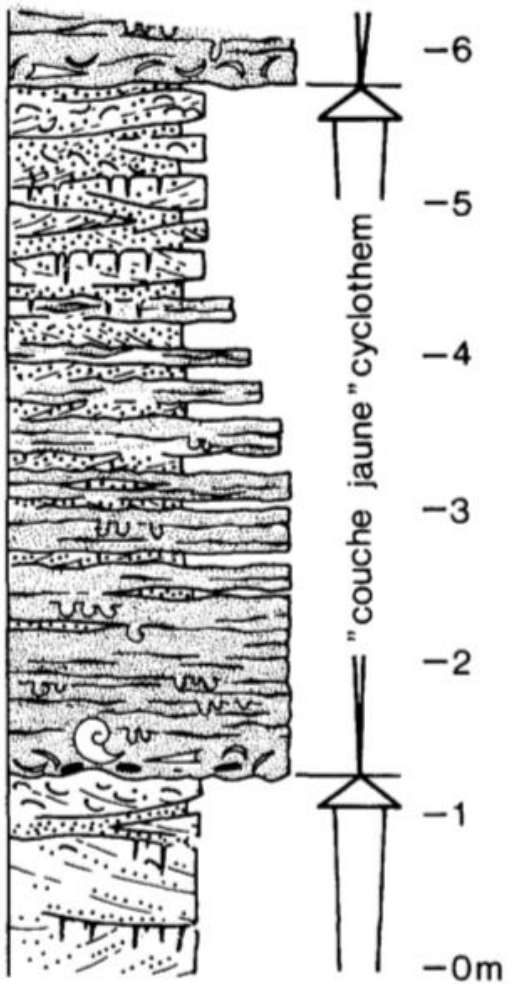
  - Hematite (Fe_2O_3)
  - Goethite ($FeOOH$)
  - Glauconite ($(K,Na)(Fe^{3+},Al,Mg)_2(Si,Al)_4O_{10}(OH)_2$
- **Common during key Phanerozoic periods: Ordovician-Silurian, Jurassic, Cretaceous**



Oolitic ironstones
Petrography



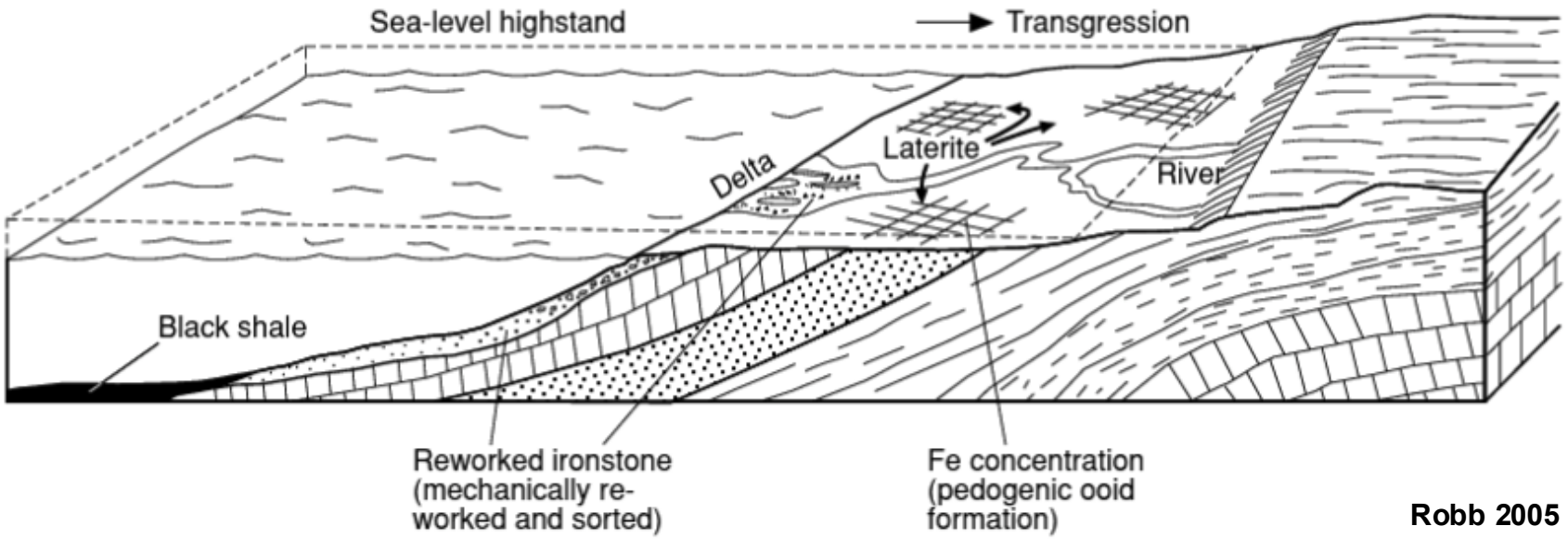




Siehl 1989

- **Evidence for shallow-water deposition: cross bedding, mudcracks, shallow-water fauna**

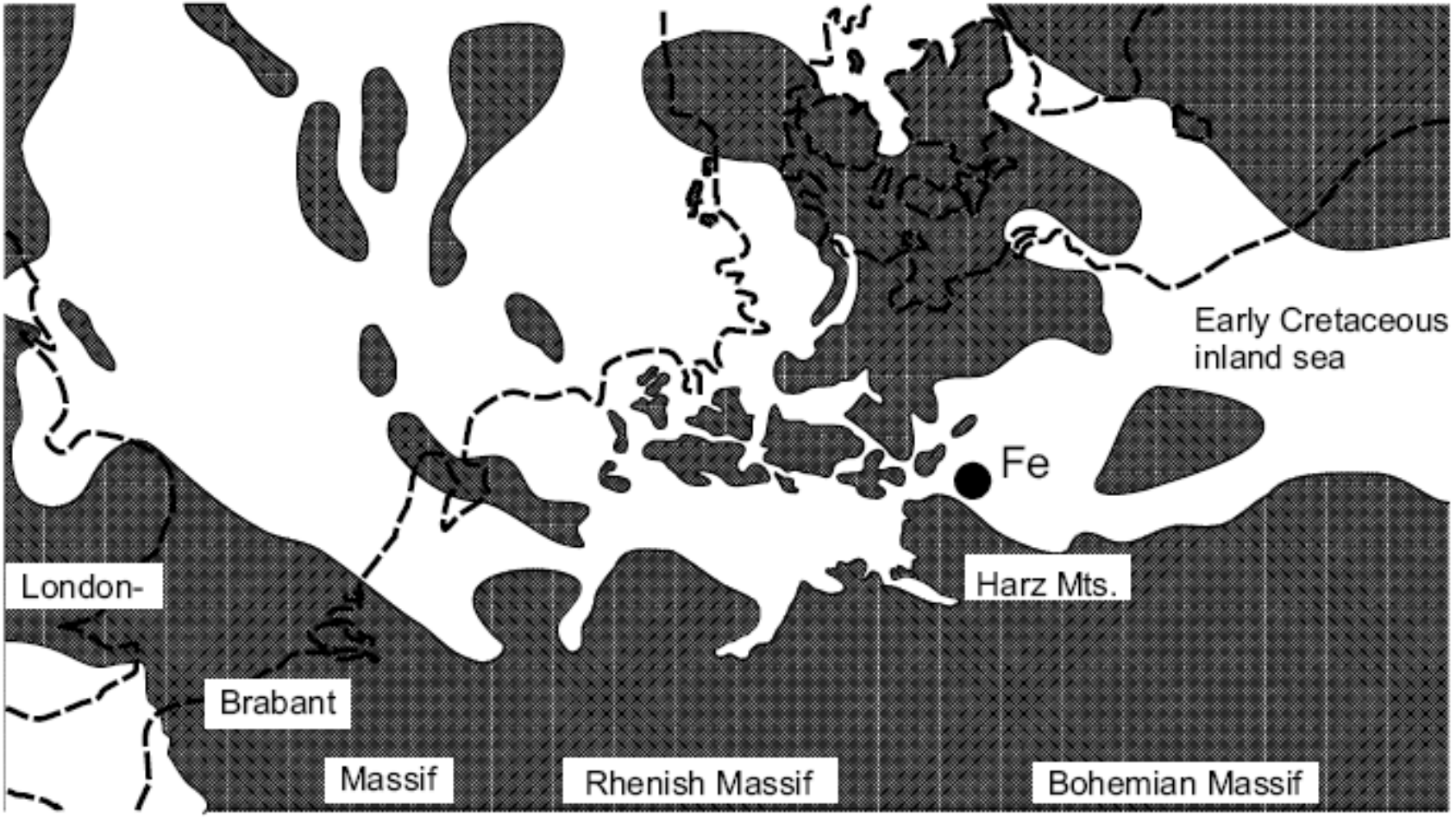
Oolitic ironstones
Genesis



Robb 2005

- Fe^{3+} sourced from continents: laterite formation in tropical climates
- In-situ formation of ooids in laterite (biochemical precipitation)
- Reworking during sea level rises/falls and deposition in high-energy shallow marine settings
- Redox changes after deposition caused mineralogical changes (e.g. goethite -> berthierine
- Linked to warm climates and global transgressions

Oolitic ironstones
Genesis



Pohl 2005

Oolitic ironstones
Mining

- Bed thickness 5cm-1m (locally <8m)
- Grades 22-38% Fe, but significant amounts of Si, Al, P
- Used as main local Fe source in 18th to early 20th century
- Locations UK: East Midlands (Northhampton) and Yorkshire (Staiths)
- Common Lorraine (France-Jurassic), Saxony (Germany-Cretacous)), Algeria (Devonian), USA (Silurian)



Warrington 2013

